

Research Article

Automation of a Pharmaceutical Batch Mixer Using Programmable Logic Controller (PLC) and Supervisory Control and Data Acquisition System (SCADA)

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Abstract

This research results from the need for an effective, reliable, and highly flexible method of control of processes within a small pharmaceutical environment and will be programmed using Ladder Logic programming Language, Siemens Step 7, which entails integrating hardware and software to provide dependable and effective operations. The mixer is controlled through a Supervisory Control and Data Acquisition (SCADA) system and programmed using Ladder Logic in Siemens Step 7 PLC Programming Software. Temperature control, agitation control, and ingredient addition sequences were all integrated by the automated system. The performance of the modeled system is validated through computer simulations. The Batch Mixer is controlled using a Variable Speed Drive (VSD), enabling precise execution of the required processes throughout the entire mixing cycle. Each phase of the process is carefully controlled and monitored, with detailed simulation results demonstrating enhanced performance and efficiency in the proposed batch mixer system, as well as offering a mathematical model of the Batch Mixer and outlining the performance variables that the PLC systems use in detail. As a backup validation tool, Rockwell Automation's RSLogix 5000 PLC is used to verify the validation. The study demonstrates the effectiveness of using Siemens STEP 7 PLC for automation and validation, and with RSLogix 5000 PLC for confirmation of validation, ensuring reliable and efficient pharmaceutical batch mixing operations.

Keywords

Automation, Batch Mixer, Human Machine Interface (HMI), Ladder Logic Diagram, Programmable Logic Controller (PLC), Supervisory Control and Data Acquisition (SCADA)

1. Introduction

Automation makes use of control systems and information technologies to reduce the need for human work in the production of goods and services. In the scope of industrialisation, automation is a step beyond mechanisation [1] (Anon., 2015a). Whereas mechanisation provided human operators with machinery to assist them with the muscular requirements of work,

automation greatly decreases the need for human sensory and mental requirements as well. An example is the Automated Teller Machine (ATM) used by the banks. Automation has had a notable impact in a wide range of industries such as the manufacturing, power and service sectors [2] (Anon., 2015b). It requires a conscious effort on the part of plant managers and

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Received: 9 August 2024; **Accepted:** 5 November 2024; **Published:** 26 November 2024



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engineers to identify the areas where automation can result in better deployment and/or utilisation of human resource and savings in man-hours or down time. Controls automation need not be high ended and extremely sophisticated; it is the phased, step-by-step effort to automate, employing control systems tailored on one's specific requirements that achieves the most attractive results. A Supervisory Control and Data Acquisition (SCADA) makes industrial automation easier to understand and troubleshoot. Batch production enables the manufacturing technique of creating a group of components at a particular production plant before moving the group to the next step in production. This technique is common in ice cream mixes, manufacture of fertilizer and animal feed, flour and paint mixes. According to [3] Kumar (2015) operator-based monitoring of batch mixing affects product quality, and offers inefficient production.

Automation by way of combining SCADA with Programmable Logic Controller (PLC) has received much attention from researchers in both industry and academia. This is attributable to the need to address industry problems cost effectively and at same time boost product quality, production rate, machine reliability, and minimize human intervention. In this regard, PLC-SCADA systems have been successfully applied in the automation of a cement kiln mill drive to showcase accuracy in speed regulation with machine protection during variable load operation of mill [4] (Harchandani and Bintu, 2014), automation of an industrial wood pellet burner to achieve remarkable energy conservation and full plant control [5] (Pandey and Bhatt, 2015), improvement in effectiveness and efficiency of the production process of industrial machines in terms of cost and time savings [6] (Sethiawan *et al.*, 2019), precise, solid, and more efficient control of home appliances [7] (Mishra, 2020), tracking the setpoint angle position of a directional wheel conveyor to achieve desirable displacement accuracy [8] (Howimanporn *et al.*, 2021), prediction of electric power production of a hydroelectric power station generator in order to balance with the national grid in real-time [9]

(Myint *et al.*, 2021), and the continuous monitoring, control, fault detection and protection of 3-phase induction motor to drastically reduce the time and implementation costs of new control circuits.

More importantly, [10] Sanamdikar and Vartak (2013) developed a system for colour making and mixing using Allen Bradley Micrologix 1000PLC and SCADA. Desired ratios of the three 36 main colours were utilised to create nine colours with different proportions so it will be used in different petrochemical companies. [11] Okoli *et al.* (2013) used the Logix pro 500TM PLC to simulate control of an industrial mixer. Properties of the mixer were modelled using Virtual Lab software that provided library of parameterised modules representing the actual behaviour of the mixer components. To pump water from an underground tank to an overhead tank automatically, [12] Das *et al.* (2013) implemented an automated tank level control system with the help of SIMATIC S7-300 PLC and SCADA Human Machine Interface (HMI). [13] Dhiman and Kumar (2014) proposed the design, development and testing of a prototype of commercial hybrid method of automatic bottle filling using PLC and SCADA. The purpose was to apply the design to filling two types of chemical liquids into bottles randomly by using PLC as a controller for a batch operation. [14] Aung and Aung (2019) achieved high throughput, improved quality and consistency of product by automating a non-stop, on-conveyor grains filling process by the application of Siemens Simatic WinCC Flexible Advanced Software HMI and Siemens Step 7 PLC ladder diagram. [15] Tomar and Kumar (2020) successfully automated bottle-sorting, liquid filling, capping, labelling, and counting stages of a filling process in a beverage industry using Wonderware InTouch SCADA and OMRON PLC software, inductive proximity sensors as well as photoelectric retroreflective sensors. Also, a 3-tank bottle filling and capping fluid or chemical blending system using a SCADA HMI interface and Mitsubishi GOC35 PLC developed by [16] Agarwal *et al.* (2020) to minimise human intervention and achieve higher product quality.



Figure 1. The Batch Mixer to be Automated.



Figure 2. Auxiliary Tanks of the Batch System.

From the reviewed literature, no work has been reported on PLC- SCADA automation of a pharmaceutical batch mixer. This paper contributes by enhancing knowledge on SCADA automation of processes by giving focus to the automation of a manually operated batch mixer that exhibits the following:

- 1) Manual haulage of sugar crystals into the heating tank;
- 2) Manual ON-OFF operation of valves; and
- 3) Lack of a decision support system.

The rest of the paper is organised as follows: In Section 2, we present the process description and the design of PLC and SCADA programs. The PLC simulation results and SCADA interface are presented in Section 3, and Section 4 and 5 discusses the results and conclusion respectively.

2. Materials and Methods

2.1. Design Concept and Criteria

The design concept is shown in Figure 3 as the proposed automated batch mixing system. The design criteria for the proposed batch mixer are as follows:

- 1) The system should minimise spillages and human errors;
- 2) Information on real-time plant performance should be made available to Management;
- 3) Ingredients to be mixed should be readily available to the batch mixing system; and more so there should be automatic haulage of sugar crystals;
- 4) Operation of valves and detection of level of material in the tank should be automatic; and
- 5) Hard wired relay-contactor controls should be replaced with soft wired versions for efficient production

2.1.1. Process Description

The batch mixing processes are of both time and event driven operations. The following statement list for a reaction and blending process for which the process timing diagram is constructed is given as follows:

Step 1: Prepare for batch mixing process for a nominal time of 10 mins

- a. Operator washes vessels
- b. Operator inspects washed vessels for debris

c. Ingredients are weighed

d. Operator waits for START PB to close.

Step 2: Charge the heating vessel with sugar crystals for a nominal time of 3 mins

- a. Initiated when operator presses START PB
- b. The sugar crystals are sucked into the heating vessel.

Step 3: Initiate reaction for a nominal time of 45 mins

- a. Temperature controller set to MANUAL
- b. Additive added to sugar content and melted at a required temperature of 75 °C
- c. Cooling water ON after heating solution for 45 minutes
- d. Temperature then drops to 40 °C
- e. Stirrer motor turned ON and stays ON until beginning of the drain operation.

Step 4: Blend the sugar solution with other added additives for a nominal time of 60 mins

- a. Initiated after the start of reaction operation
- b. Ingredients are added
- c. Mix until homogeneous at a temperature of 75 °C.

Step 5: Drain heating vessel for a nominal time of 45 mins

- a. Initiated when temperature falls to 40 °C
- b. Cooling water valve CLOSES
- c. Filter and Pump ON
- d. Mixing vessels stirrer motor turns ON
- e. Terminate stirring when heating vessel level goes empty
- f. Heating vessel stirrer motor STOPS when vessel is empty.

Step 6: Blend the sugar solution containing additives with preservatives for a nominal time of 60 mins

- a. Ingredients are added
- b. Valve OPENS to add distilled water
- c. Mix contents until homogeneous
- d. Distilled water valve opens.

Step 7: Drain resultant contents (i.e. syrup) of mixing vessels to the storage vessels for a nominal time of 20 mins

- a. Mixing vessel stirrer motor stays ON
- b. Outlet valve opens
- c. Pump ON
- d. Storing vessel mixer motor turns ON
- e. Terminated when the mixing vessel goes empty
- f. Mixing vessel stirrer motor turned OFF.

Step 8: Cool the syrup

a. Mixer motor stirs to attain room temperature

b. Ready for bottling

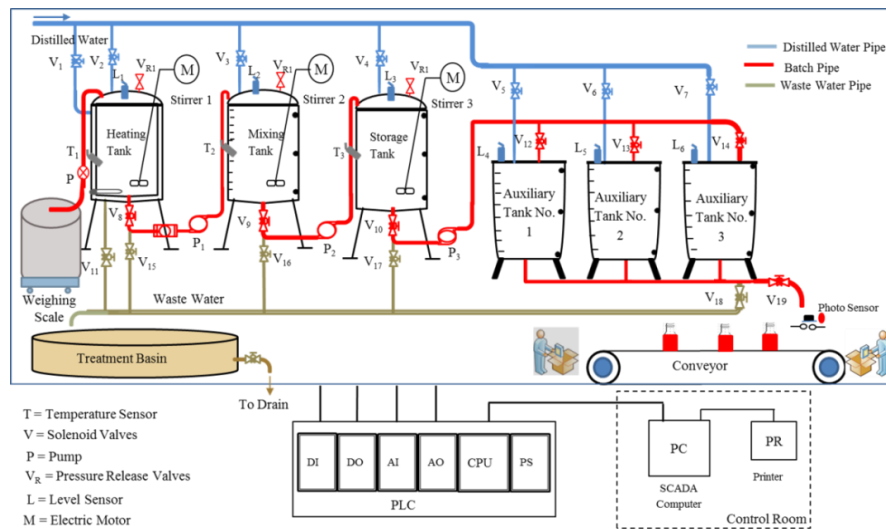


Figure 3. Proposed Automated Batch Mixing System.

2.1.2. Mathematical Modelling

The system to be modelled is presented in Figure 4. The modelling of the batch mixer system is reorganised into three sections as shown in Figure 4 and figure 5.

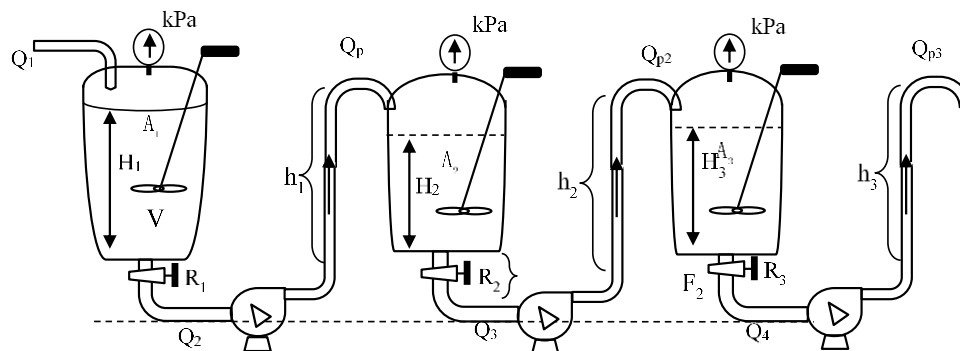


Figure 4. Batch Mixing System to be Modelled.

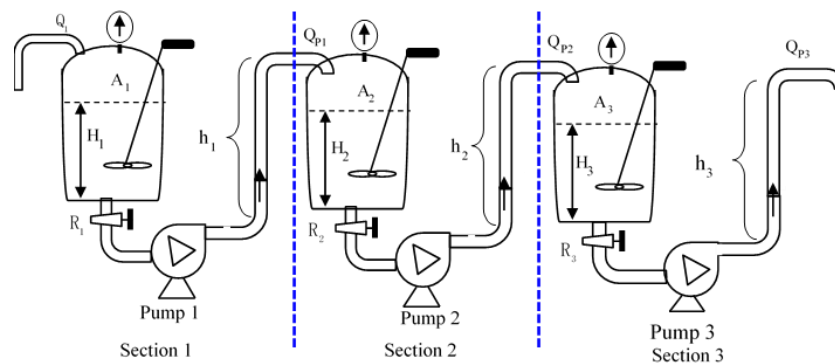


Figure 5. Sectionalisation of the Batch Mixing System to be Modelled.

For the modelling, the following equations were utilised [17, 18]: Equation (1) and Equation (2) are the mass flow balance equations, Equation (3) is the Bernoulli's energy balance equation, Equation (4) gives the efficiency of pump, Equation (5) gives the volumetric flow rate through a pump, Equation (6) expresses the work done by the pump and Equation (7) is Darcy's equation which expresses the frictional head loss (caused by eg. friction, valve expansion and bends) through the pipe.

$$m'_i = m'_e + \frac{dM}{dt} \quad (1)$$

$$Q_i = Q_{i+1} + A_i \frac{dH_i}{dt} \quad (2)$$

$$P_i + \rho gh_i + \frac{1}{2} \rho V_i^2 + h_A - h_E - h_L = P_{i+1} + \rho gh_{i+1} + \frac{1}{2} \rho V_{i+1}^2 \quad (3)$$

$$\eta_i = \frac{P_{i \text{ out}}}{P_{i \text{ input}}} \quad (4)$$

$$Q_{Pi} = \frac{P_{i \text{ out}}}{h_A \gamma} \quad (5)$$

$$h_A = \frac{P_2 - P_1}{\gamma} + \frac{V_2^2 - V_1^2}{2g} + h_L \quad (6)$$

$$h_L = \frac{\Delta P}{\gamma} = \frac{P_1 - P_2}{\gamma} = \left(\frac{4\tau_w}{\rho g} \right) \left(\frac{L}{D} \right) = f \left(\frac{L}{D} \right) \left(\frac{V^2}{2g} \right) \quad (7)$$

The restriction of flow by the valve is given by Equation (8) [18].

$$R_i = \frac{H_i(t) - H_{i+1}(t)}{Q_i} \quad (8)$$

Laplace transforms applied to the volumetric flow rate and the restrictions of the valve are given by Equation (9) and Equation (10) respectively.

$$Q_i(s) = Q_{i+1}(s) + A_i s H_i(s) \quad (9)$$

$$Q_i(s) = \frac{H_i(s) - H_{i+1}(s)}{R_i} \quad (10)$$

Substituting Equation (9) into Equation (10) gives the system transfer function as shown in Equation (11).

$$\frac{Q_{i+1}(s)}{Q_i(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{1}{A_{i+1}R_{i+1}s + 1} \quad (11)$$

where, m'_i initial mass flow rate, kg/s = volumetric flow rate ρQ_i , (m³/s)

m'_e = mass flow rate at exit, kg/s = volumetric flow rate ρQ_{i+1} , (m³/s)

M = mass of the system, kg/s = volumetric flow rate $\rho A_i H_i$, (m³/s)

γ = specific weight of fluid moving in the pipe, (dimensionless quantity)

h_A = pump work done, (m)

h_E = loss through extraction (e.g. turbine and windmill) = 0, (m)

h_L = frictional head loss through pipes (e.g. friction, valve, expansion and bends), (m)

V_1 and V_2 = velocity at entry and exit, (m/s)

η = efficiency of pump P_i , (no unit)

P_{iout} = flow power delivered by the pump, (Watts)

P_{input} = power input by the motor. (Watts)

Q_{Pi} = volumetric flow rate of pump (m³/s)

f = frictional factor, (no units)

L = length of the pipe, (m)

D = inner diameter of the pipe, (m)

P_1 and P_2 = pressure at the inlet and outlet of the pipe, (Pa, N/m²)

v = mean velocity (m/s)

g = acceleration due to gravity (m/s²)

ρ = density of the fluid, (kg/m³)

G = gain of the system (dimensionless)

H = feedback gain (dimensionless)

R = valve resistance, (sec/m²)

τ_i = time constant, (sec)

H_i = height of the tank (m)

A_i = area of tank (m²)

h_i = height of pipe (m)

τ_w = wall shear stress (Pa)

Equations (1) to Equation (11) are then applied to all the three sections.

2.2. Design of the PLC Ladder Logic Program for the Batch Mixer

2.2.1. Systems Flowchart

A flow chart in Figure 6A and 6B generated from the automation of the batch mixer describes the PLC controlled sequence of various components and aid the PLC programming which gives a summary of how data flows in the system and how decisions are made by the PLC to control various events.

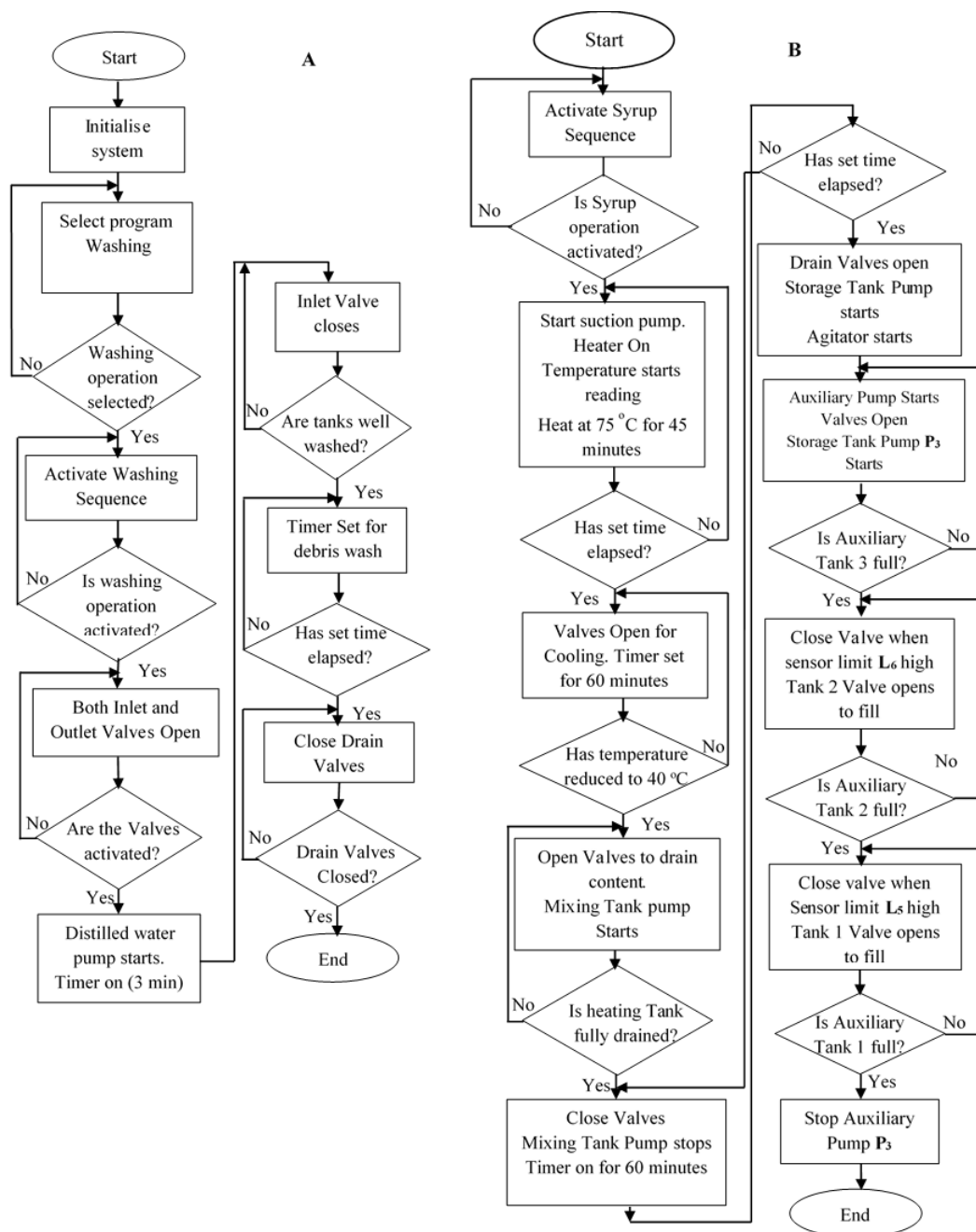


Figure 6. Flowchart for Washing and the Batch Mixing process.

2.2.2. Programming of the PLC Using the Siemens S7-300 TIA Portal PLC Software

Siemens S7-300 TIA Portal was used to develop the ladder diagram programming languages used for the Batch Mixing System. The Siemens S7-300 TIA Portal includes components like the SIMATIC manager, symbol editor, hardware

configuration, and hardware diagnostics. The integral package of the Siemens S7-300 TIA Portal is the programming languages which includes Ladder Logic, Statement List, and the Function Block Diagram. Each rungs input and output are properly addressed. Figure 7 through to Figure 9 depict the batch mixer's programming for the heating, mixing, and storing processes, respectively.

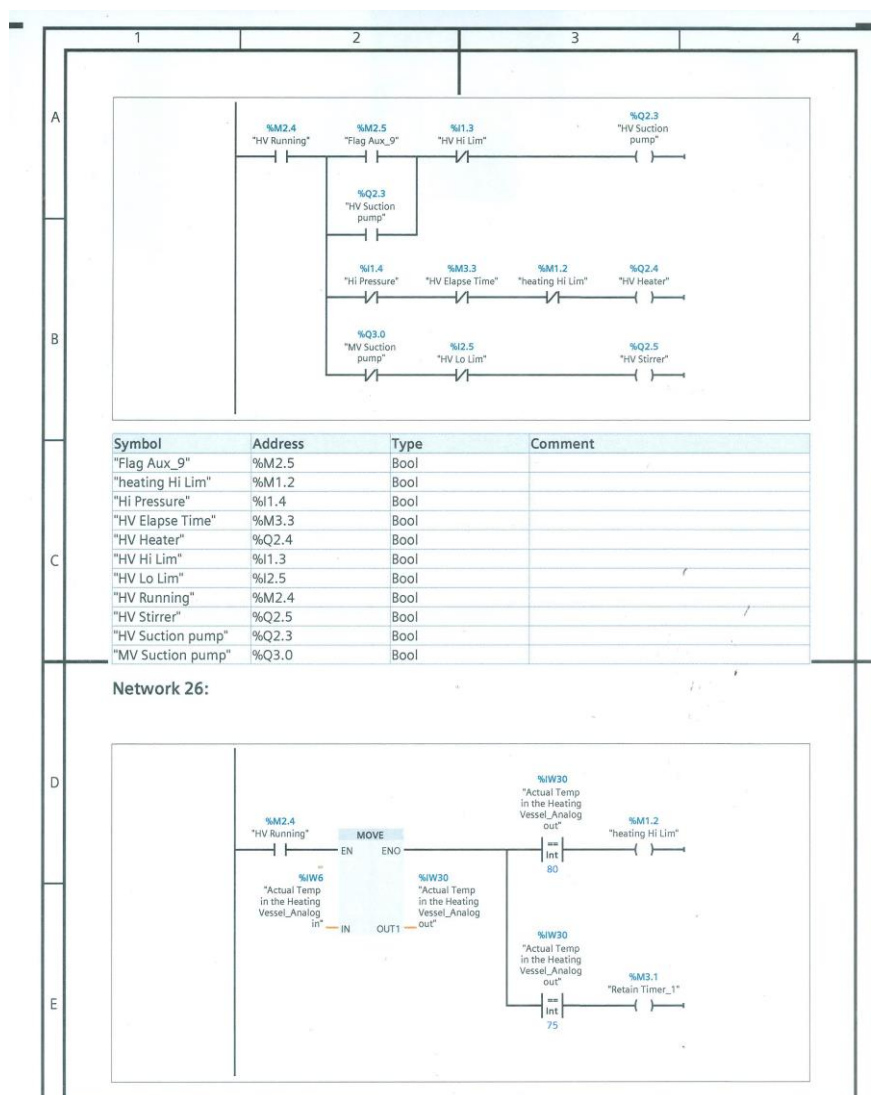


Figure 7. Suction and Heating Program for the Batch Mixing System.

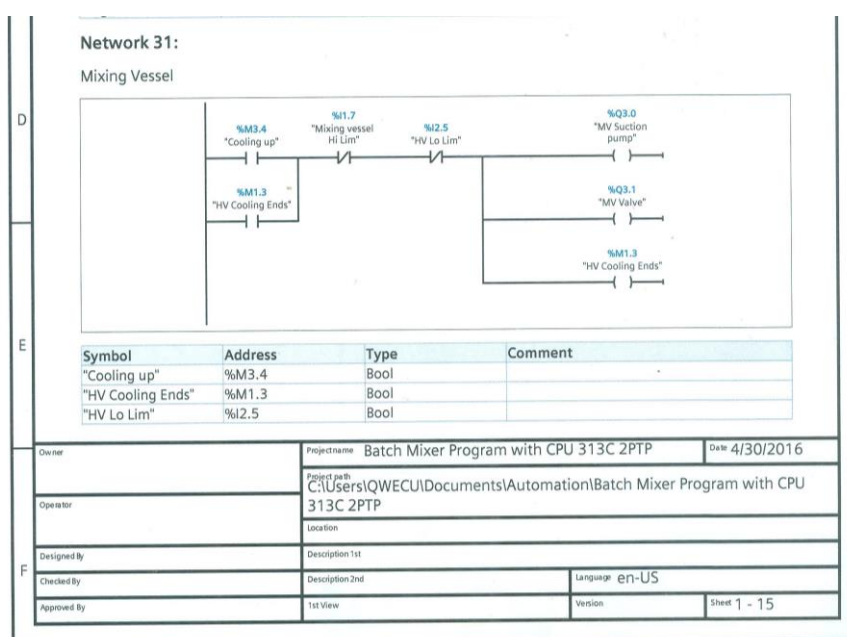


Figure 8. Program for Mixing of Syrup in the Tank.

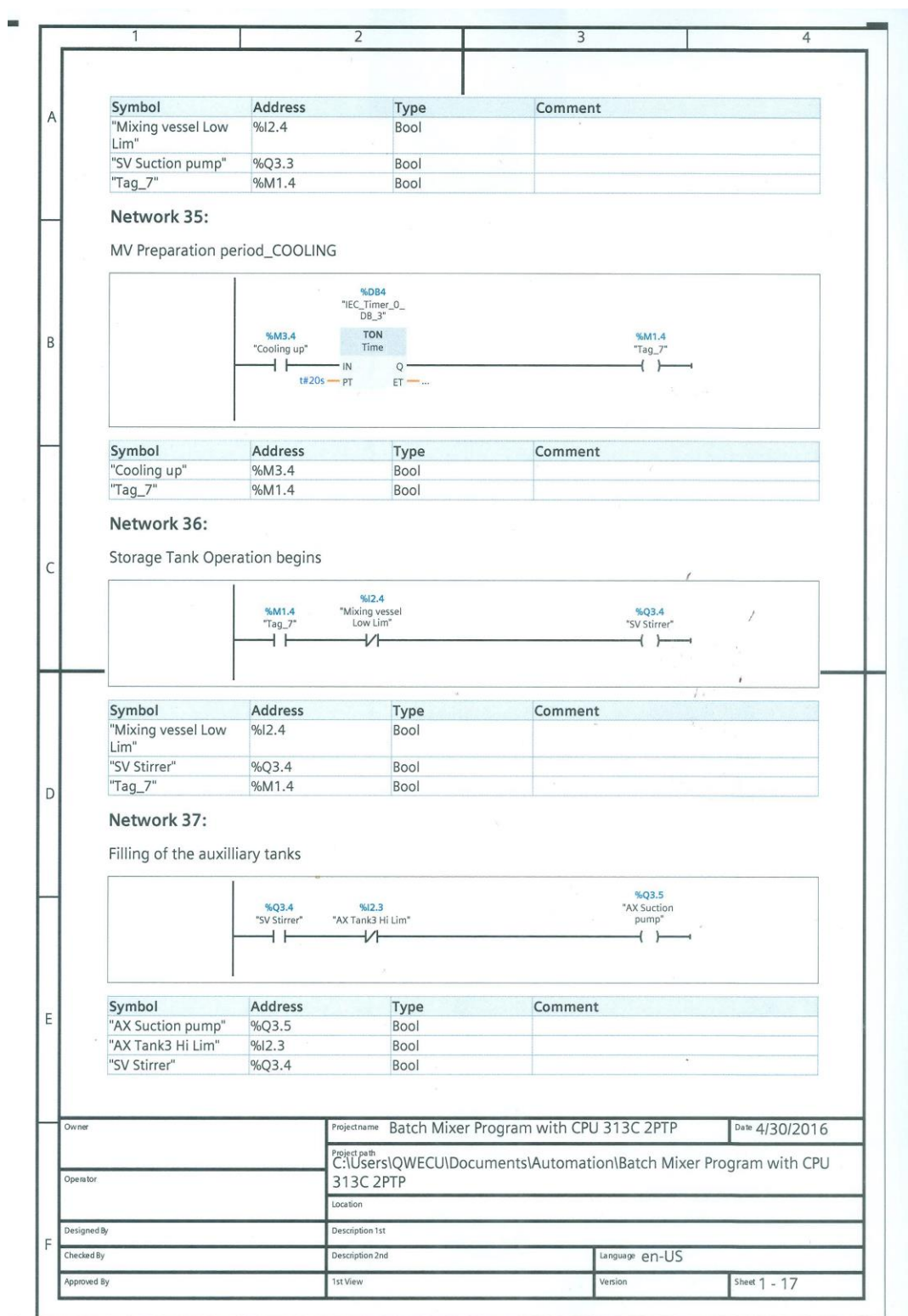


Figure 9. Program for Storage of Syrup in the Tank.

2.2.3. Programming of the PLC using RS Logix

5000 PLC Software to confirm the validation

As a backup validation tool, Rockwell Automation's RS Logix 5000 PLC was used to corroborate the validation

results from the Siemens STEP 7 PLC. This stage guaranteed the automation system's correctness and dependability. Confirmation data from RS Logix 5000 Emulate, was utilized to summarize the mixing procedures, was compared to validation data from Siemens Step 7 PLC. The batching process's stages of heating, mixing, and storage are shown in

Figure 10 through Figure 12 below.

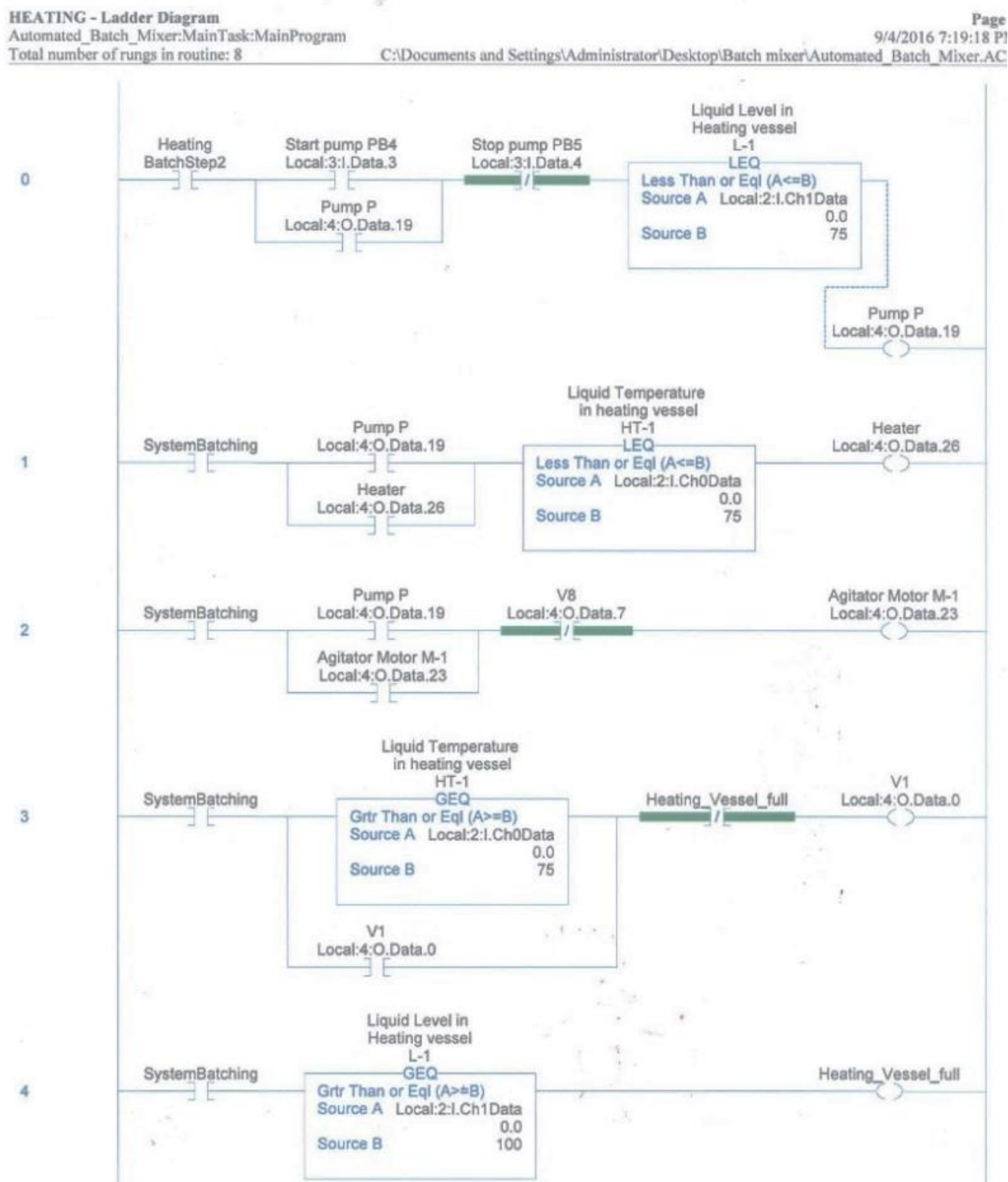


Figure 10. Heating program of the Batch Mixing System.

MIXING - Ladder Diagram

Automated Batch Mixer:MainTask:MainProgram

Total number of rungs in routine: 5

C:\Documents and Settings\Administrator\Desktop\Batch mixer\Automated Batch Mixer.A

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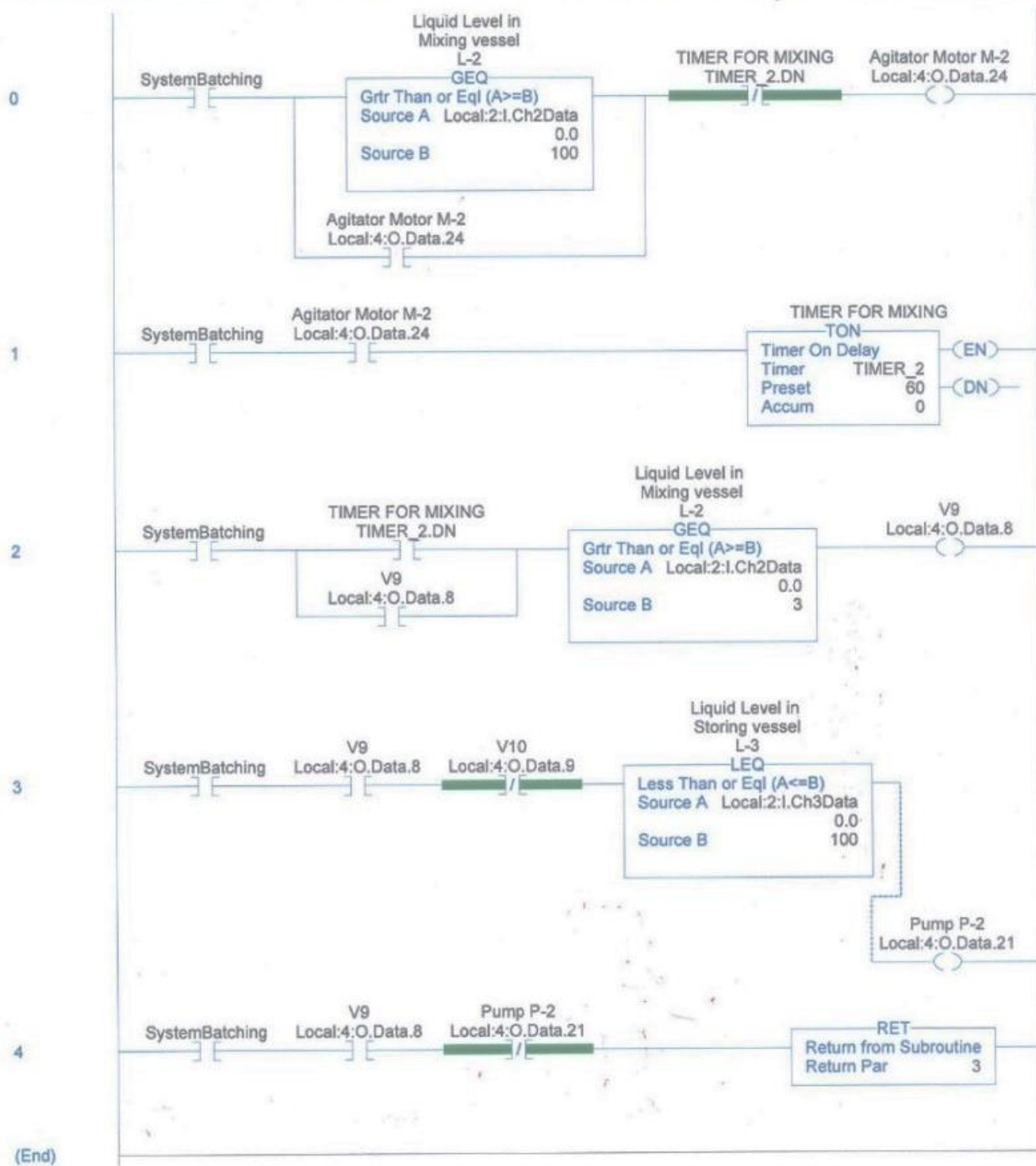


Figure 11. Mixing Program of ingredients of the System.

STORING - Ladder Diagram

Automated_Batch_Mixer:MainTask:MainProgram

Total number of rungs in routine: 8

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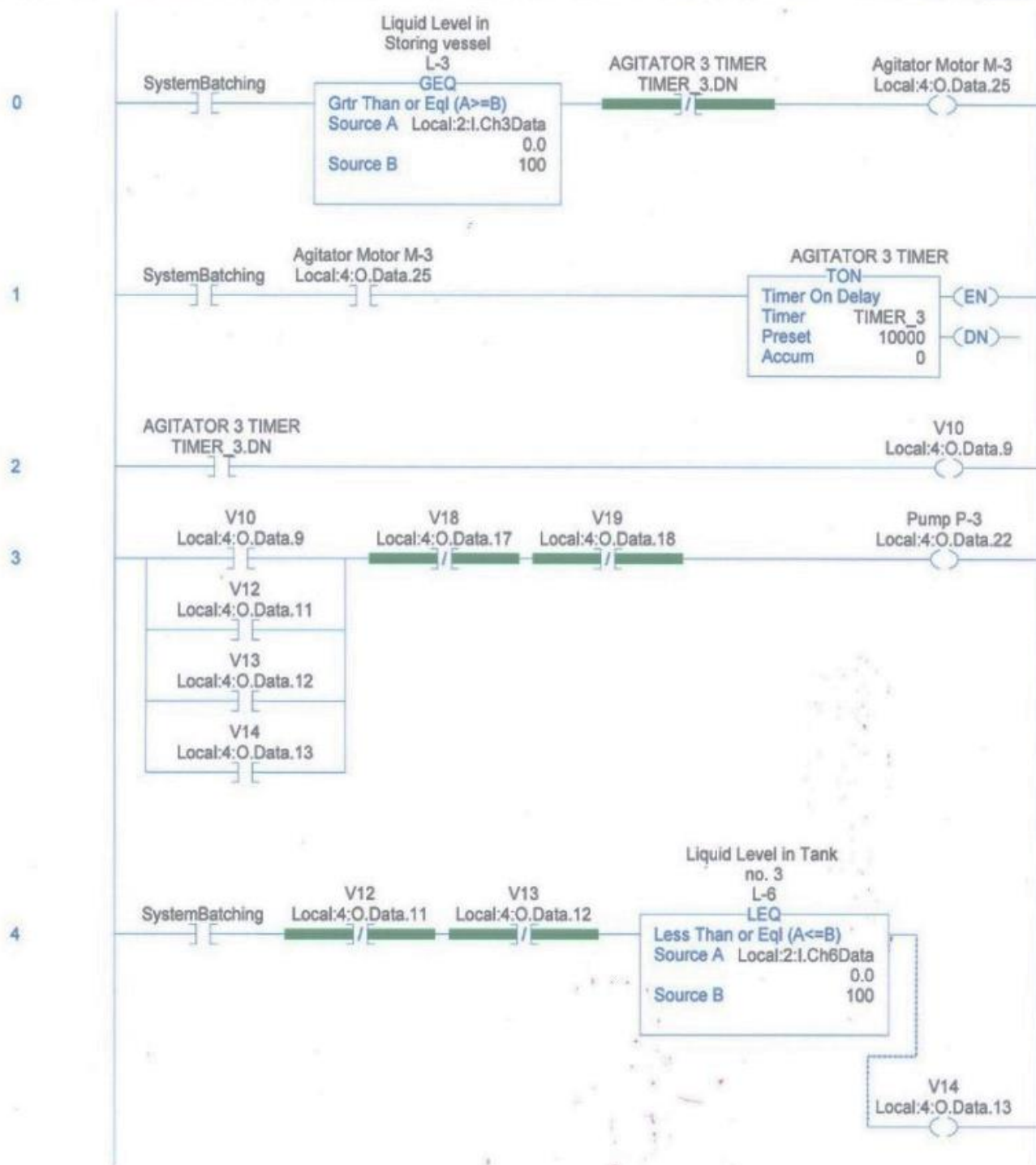


Figure 12. Storage Stage in the Batch Mixing System.

2.2.4. Design of the Supervisory Control and Data Acquisition System of Batch Mixer

The PLC automated batch mixing system is integrated into the SCADA and allow display of various plant conditions in

real time. Data is taken from remote site and compared with set points values. If set point falls within range the process continues, if system fall outside range then an alarm is displayed and operator issues a command. Figure 13 is a flowchart of actions and functionalities of the SCADA display and control alarm system.

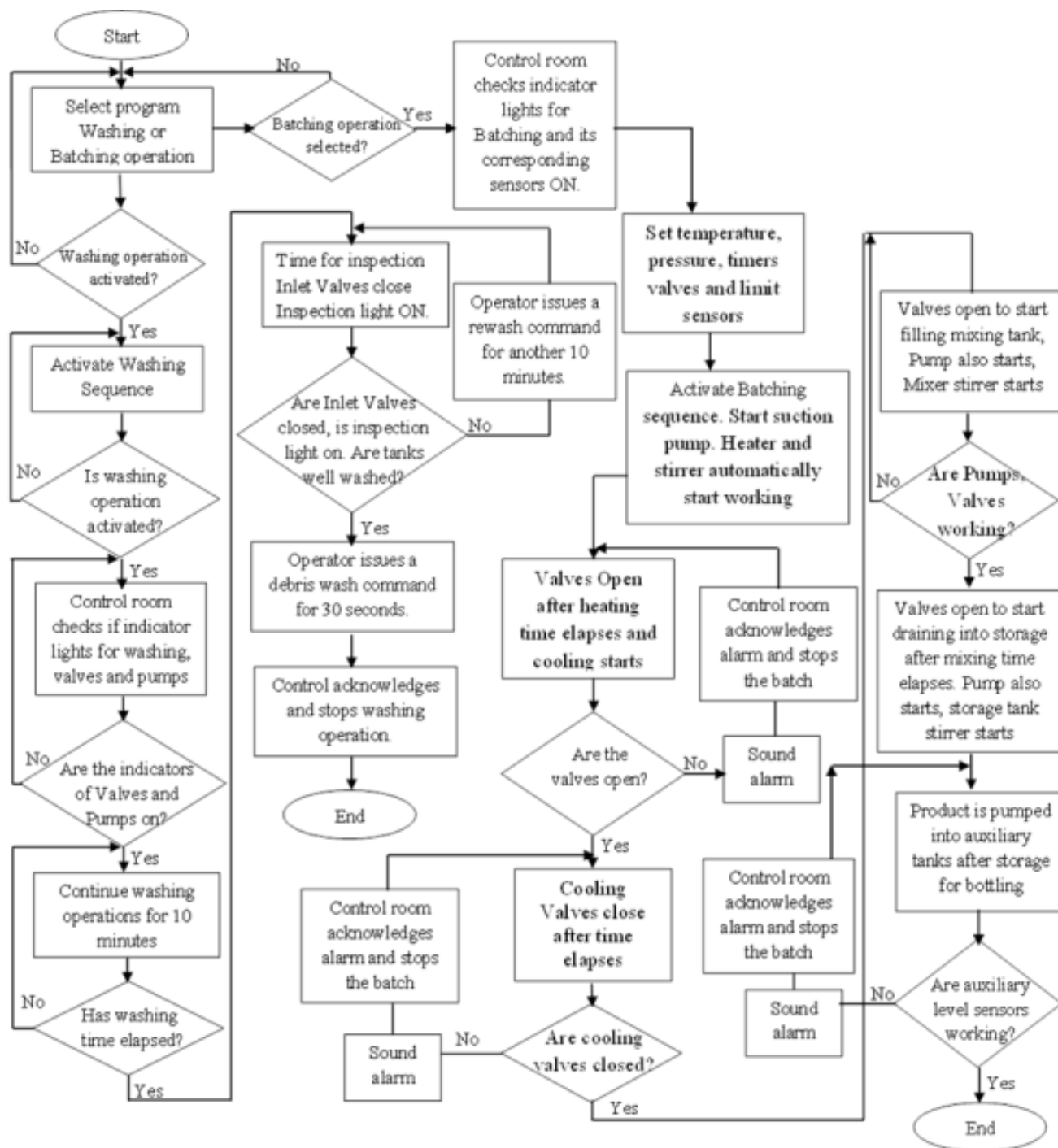


Figure 13. Flowchart of SCADA System.

2.2.5. Programming of the InTouch Wonderware SCADA HMI Software

The SCADA InTouch WindowMaker program of the software is used to design the graphic system of the various components/layouts of the batch mixer. Window script was used to write the various commands needed to simulate the project; the SCADA design enables the user/operator to visualise whatever process is going on in order to monitor and

control the plant and equipment. However, if the physical SCADA is used, the entire PLC input and output addresses shall be written in the SCADA program. Programming the SCADA starts with the HMI creation and design of the interface using the InTouch Wonderware by setting up a communication link between the HMI device and PLC using tags, user data areas and networks. The SCADA interface using InTouch-Window Maker, and components selection using the “Wizard Symbol Factory Library”.

3. Results

The findings and the inferences that can be drawn from the studies are succinctly and accurately described in the results.

3.1. Simulation Results of PLC Programming Using Ladder Diagram

The simulation results of the Siemens S7-300 TIA Portal

software, with the programming divided into two parts namely, the washing operation and the batching operation are presented.

3.1.1. Simulation Results for Siemens S7-300 TIA Portal Programming

Snapshots of the simulated washing, batching operations are shown in Figure 14 through to Figure 17. The entire simulation program could be found in Appendix A.



Figure 14. Valves Open for Washing.

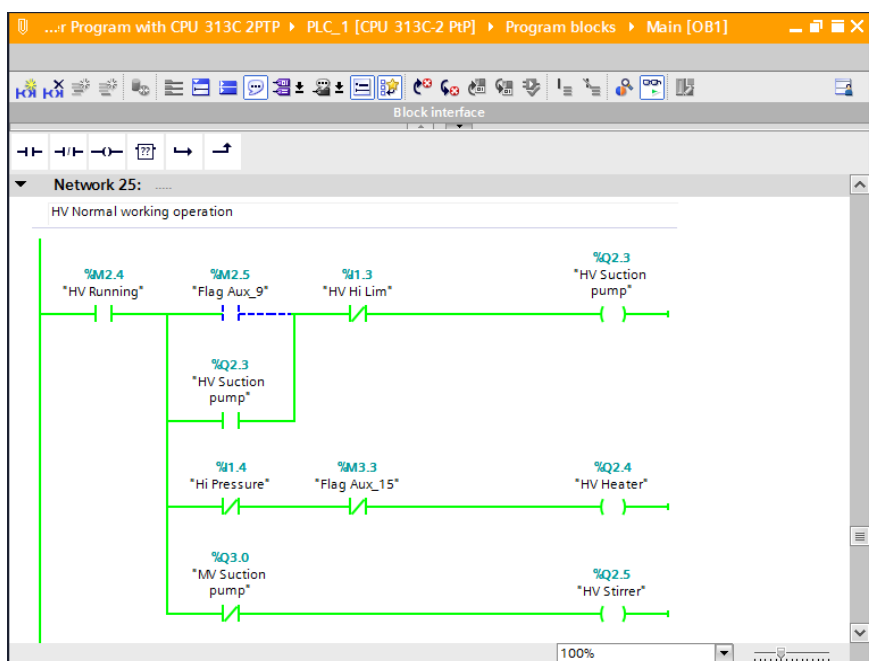


Figure 15. Heating Tank Agitator and Heaters Start.

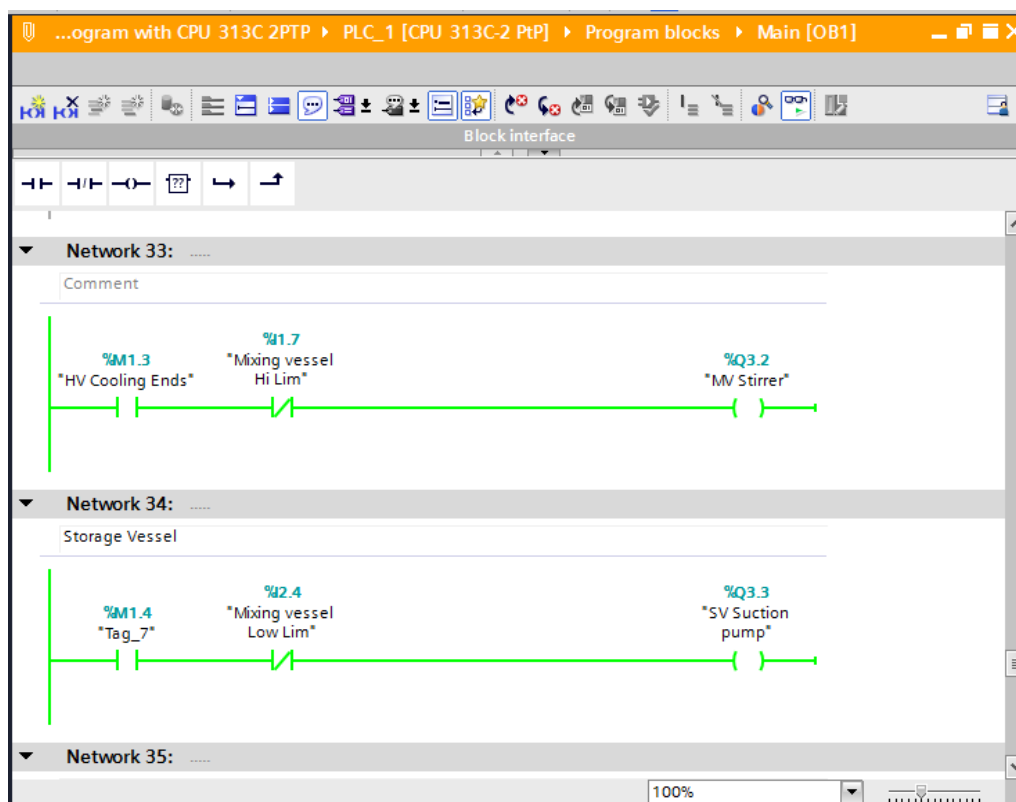


Figure 16. Mixing Tank Agitator Starts.

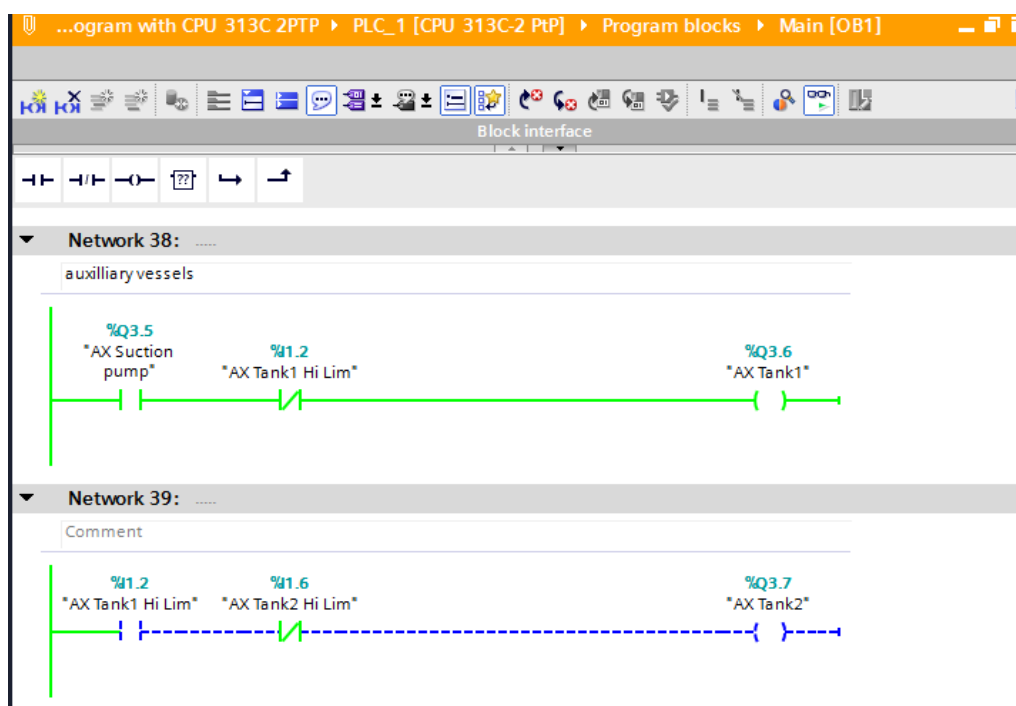


Figure 17. Filling Auxiliary Tank for Storage and Bottling.

3.1.2. Simulation Results for RS Logix 5000 Programming

Snap shots of the simulated washing, heating and storing operations are shown from Figure 18 through to Figure 20.

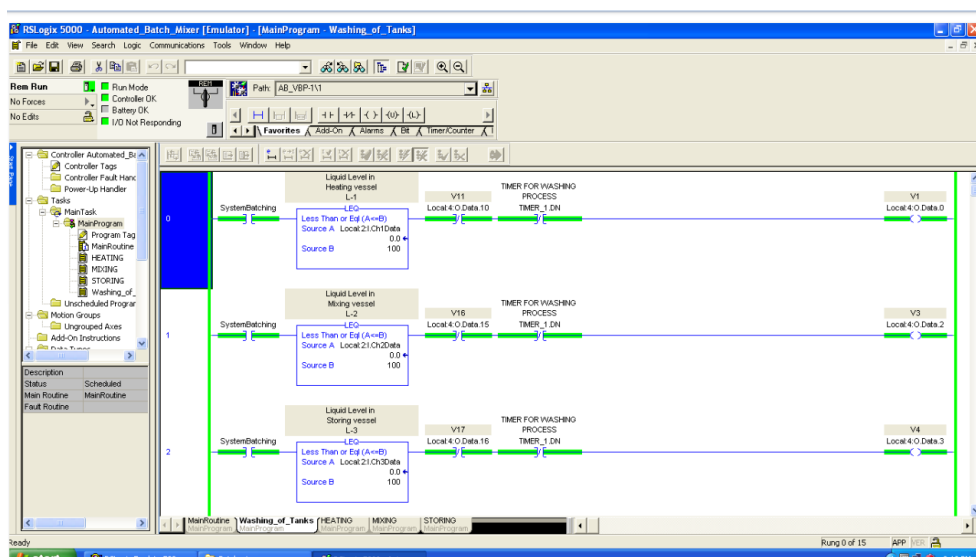


Figure 18. Simulation Results of the Washing Operations.

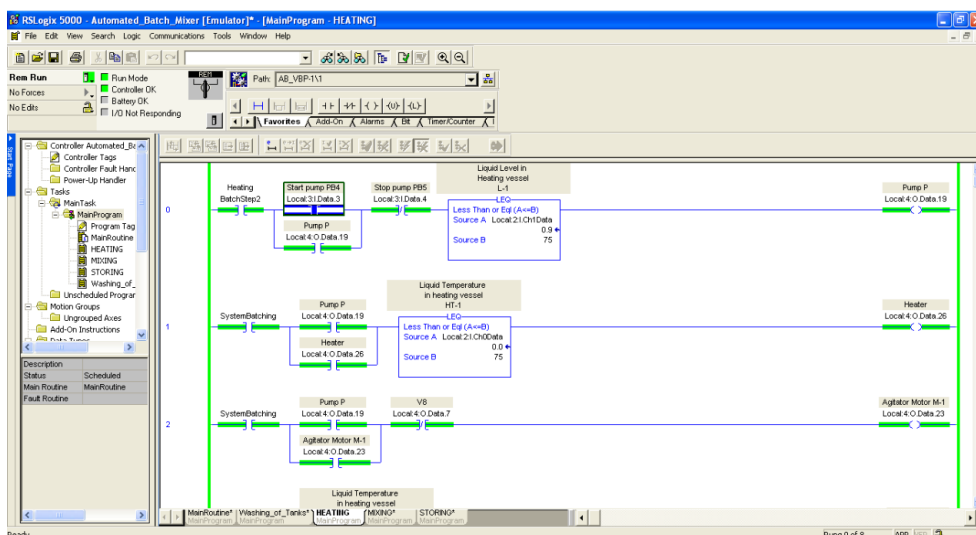


Figure 19. Simulation Results of the Heating Operations.

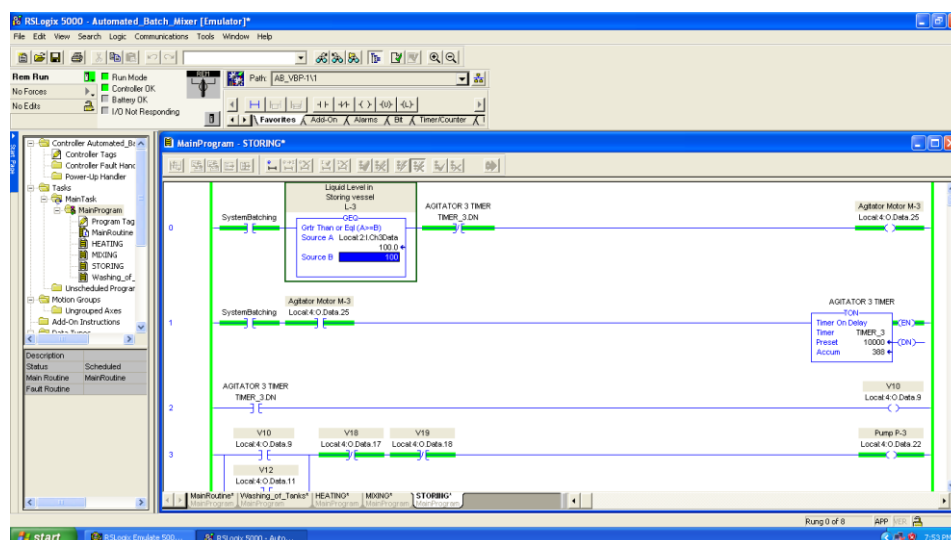


Figure 20. Simulation Results of the Storing Operations.

3.1.3. SCADA HMI Design and Implementation

The design and implementation of the Human-Machine Interface (HMI) for the automated pharmaceutical batch mixing system followed the ISA-101 standard to create an intuitive, efficient, and reliable system for operators to interact with the process. This HMI was designed to manage, monitor, and control the batch mixing operations while integrating seamlessly with the Supervisory Control and Data Acquisition (SCADA) system and the Programmable Logic Controller (PLC). The design process, key features, and implementation steps are detailed below to illustrate the systematic approach taken.

HMI Design Using ISA-101 Standard

The HMI design was guided by the ISA-101 human-centered approach, focusing on usability, task-specific screens, and ease of navigation for operators. The design incorporated a hierarchical structure to allow operators to view general information at a high level and drill down into more detailed data as needed.

Overview Display

At the top of the hierarchy, the level 1 overview display provided a holistic view of the entire batch mixing process. This screen presented critical system status indicators such as batch phase progression, real-time temperature readings, mixer speed, ingredient levels, and valve positions. A color-coded alarm banner was positioned prominently to signal any system issues, with red indicating critical alarms, yellow for warnings, and green for normal operations as shown in [Figure 22](#). Navigation buttons allowed operators to easily access more detailed process-specific screens for further analysis or intervention.

Process-Specific Displays

Moving to the level 2 process-specific displays, operators could access detailed screens dedicated to different stages of the batch mixing process, such as heating, mixing, and storage. For instance, the heating system display allowed operators to monitor and adjust the temperature and control the cooling system once the desired temperature was reached. Similarly, the mixing system display provided real-time information on mixer speed, ingredient addition status, and homogeneity levels, allowing precise control over the blending process. The storage system display monitored syrup transfer, pump status, and valve positions, ensuring that all components of the batch were handled correctly. These detailed displays provided operators with the necessary tools to control the process more efficiently while maintaining situational awareness.

Diagnostic Displays

The level 3 diagnostic displays offered a deeper level of detail for maintenance personnel or troubleshooting. These screens provided data on pump performance, sensor readings for temperature and flow rates, and valve statuses. Historical data and trends were also made available to help diagnose issues and optimize future processes. A detailed alarm log was included to track all system alerts, including timestamps, causes, and operator responses. This diagnostic level ensured

that any anomalies could be identified and addressed quickly, minimizing downtime and improving system reliability.

Layout and Graphics

In terms of layout and graphics, the HMI followed ISA-101 guidelines to maintain a clean and consistent design across all levels. Colors were used sparingly, with green representing normal operations, yellow for warnings, and red for critical alarms, ensuring operators could easily interpret system statuses. Standard symbols for tanks, pumps, and valves were used to provide a familiar interface and minimize confusion. The overall navigation system was designed to be simple, with breadcrumb navigation helping operators track their location within the system and easily return to higher-level screens.

Alarm Management

Alarm management was a critical component of the HMI, adhering to both ISA-101 and ISA-18.2 standards. The alarm system was designed to prioritize alerts, with critical alarms receiving more prominent visual and audible notifications. Operators could acknowledge and clear alarms, but critical issues required confirmation before being dismissed. Additionally, all alarms were logged for later review, aiding in troubleshooting and process optimization. This structured approach to alarm management helped reduce alarm fatigue and ensured operators remained focused on the most pressing issues.

HMI Implementation

The HMI system was implemented using InTouch Wonderware SCADA software. The interface was designed to allow real-time communication between the PLC and the HMI, ensuring that operators received up-to-date information on the process status. The system included touchscreen controls, which enabled operators to adjust parameters such as temperature, motor speeds, and valve positions directly through the interface. Graphical trends and charts displayed real-time data for key process variables, helping operators identify any irregularities or deviations during operation. The integration with SCADA allowed for comprehensive data logging and remote monitoring, further enhancing the system's functionality.

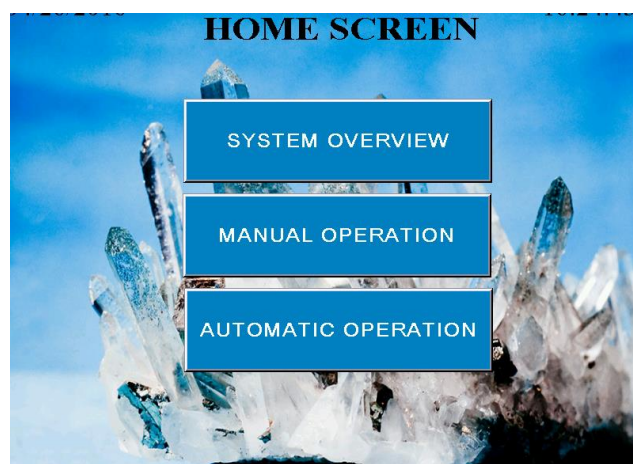


Figure 21. Interface of SCADA HMI.

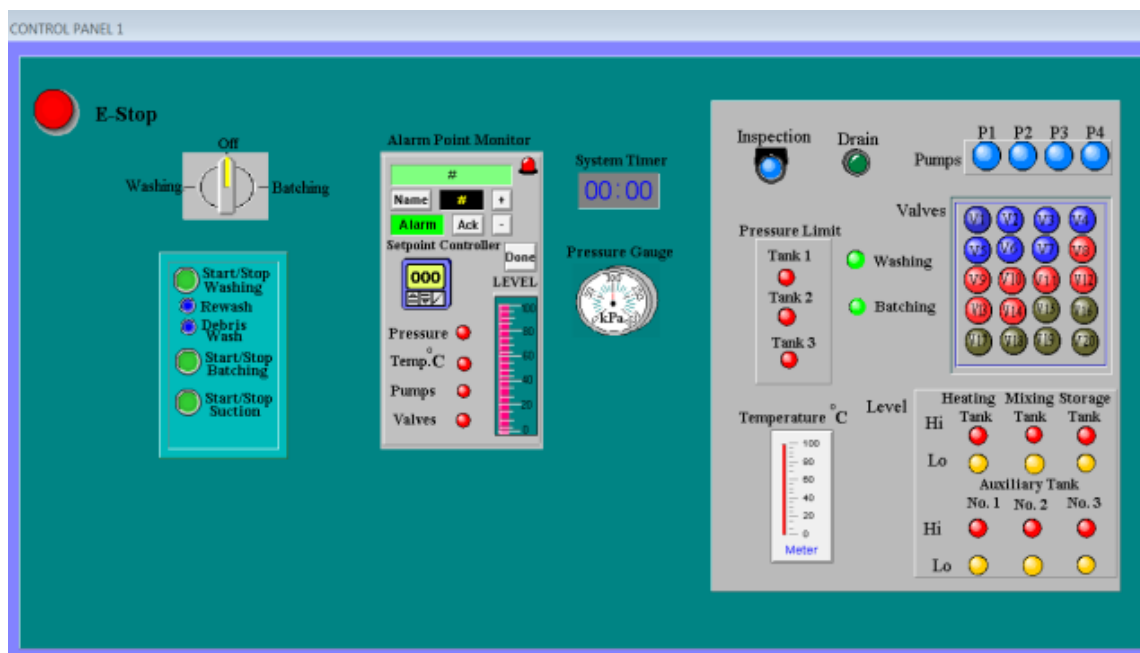


Figure 22. Interface of SCADA HMI Control Panel.

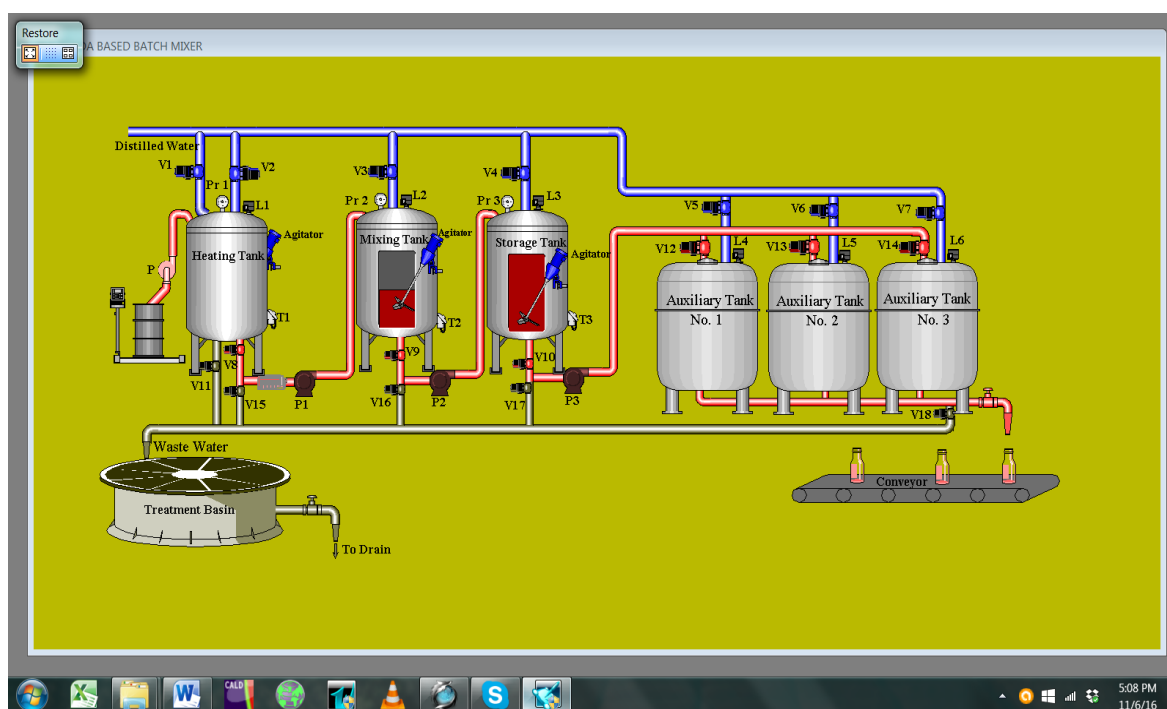


Figure 23. SCADA HMI Interface View of the Batch Mixer System.

4. Discussion

Discussion of Simulation Results

Saraswathi [19] created an automation system for a mixing process that may be applied in the pharmaceutical, food, chemical, and process industries as well as in oil refineries.

The entire suggested system is controlled by a Programmable Logic Controller (PLC). Ladder Diagrams, which are very user-friendly and simple to grasp, are used to program the MicroLogix 500 Rockwell Automation software. With the aid of a stirrer, the system automatically mixes the solutions from two separate tanks batch by batch in the third tank.

In contrast the automation project utilized Siemens S7-300 TIA Portal for simulation validation, which ensured reliability

and accuracy by comparing with real-world data, testing, and performance. High-performance computing and model simplification helped to overcome issues with data integration, system complexity.

The batch system discusses the complete batch production process, including heating, mixing, storage, and bottling. The accuracy of the simulation model was demonstrated by the RS Logix 5000 Emulate, which was used to validate the results. It showed an average variation of less than 5% and a correlation coefficient of 95%.

A 20 percent boost in predicted output, which led to an additional annual revenue increment, was another notable financial benefit of the method. Increased safety and a 20% decrease in accident and defect rates are further advantages.

To support different batch mixer installations, the ladder logic software's I/O mapping, timing, control logic, recipe management, and alarm/notification settings must be modified.

5. Conclusions

Automating a mixing or blending operation does not only lessen operator error and variations; ultimately, it increases overall process efficiency in multiple ways. The Batch Mixer System was successfully automated using Siemens S7-300 TIA Portal and InTouch Wonderware SCADA HMI Software's. The simulation results were validated using the Siemens S7-300 TIA Portal software.

This work also demonstrates that PLC is a better solution to the current batch mixer operated at the pharmaceutical company. As for reliability, development time and system per-

formance concern, PLC is better compared to relay logic implementation.

The automation of the PLC-SCADA based batch mixer provides safe and reliable system for the batch mixing process, which will increase the total production output thereby yielding significant financial benefits and savings.

The ladder logic program can easily be used to control any similar Batch Mixer Setup with some form of modification in the program.

Abbreviations

ATM	Automated Teller Machine
SCADA	Supervisory Control and Data Acquisition System
PLC	Programmable Logic Controller
HMI	Human Machine Interface

Author Contributions

Augustine Kweku Mbeah: Conceptualization, Methodology, Project administration, Software, Writing – original draft

Erwin Normanyo: Methodology, Supervision, Writing – review & editing

Conflicts of Interest

The authors declare no conflicts of interest.

Appendix

PLC LADDER LOGIC DIAGRAMS FOR THE PROPOSED BATCH MIXER SYSTEM

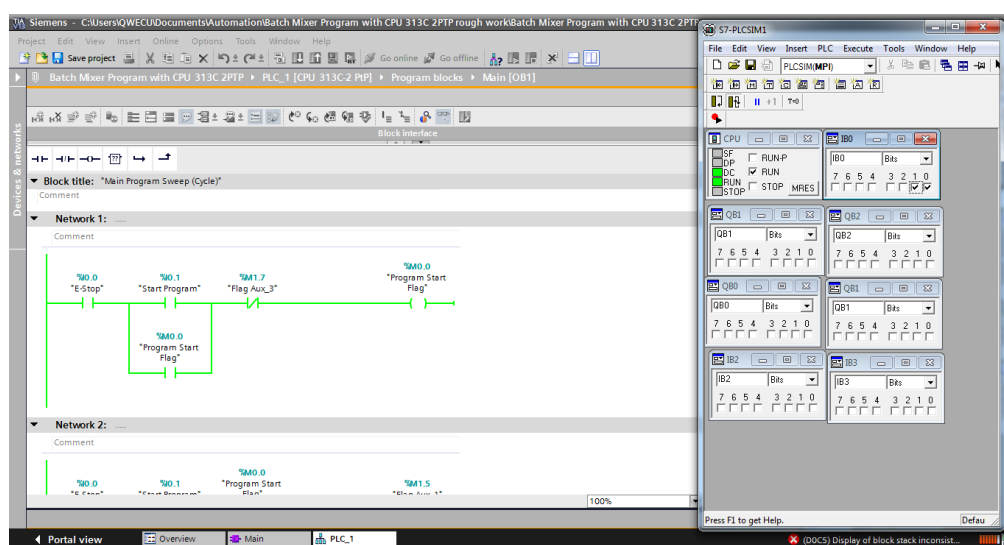


Figure A1. Starting the PLC Program.

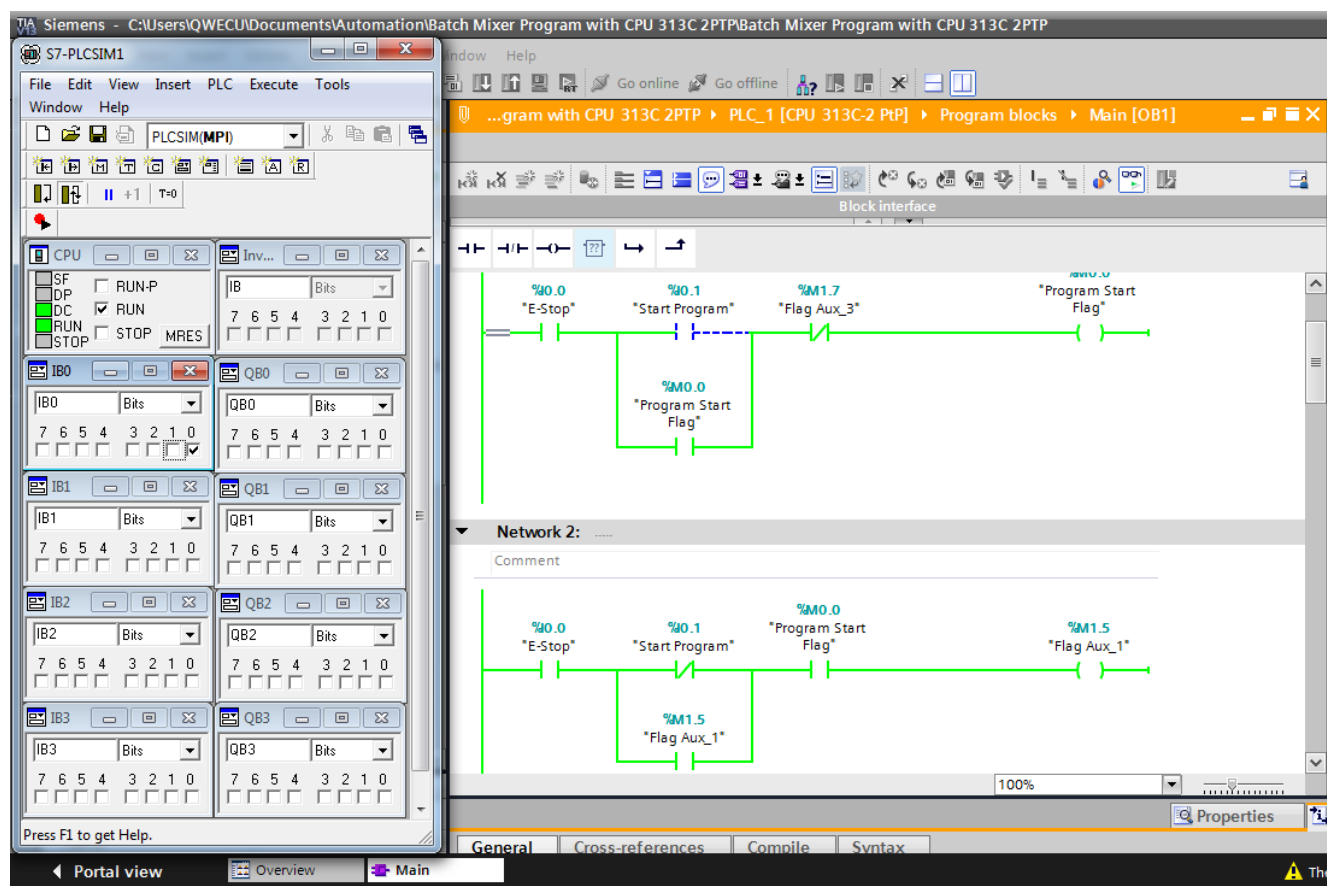


Figure A2. Program Start Flag On.

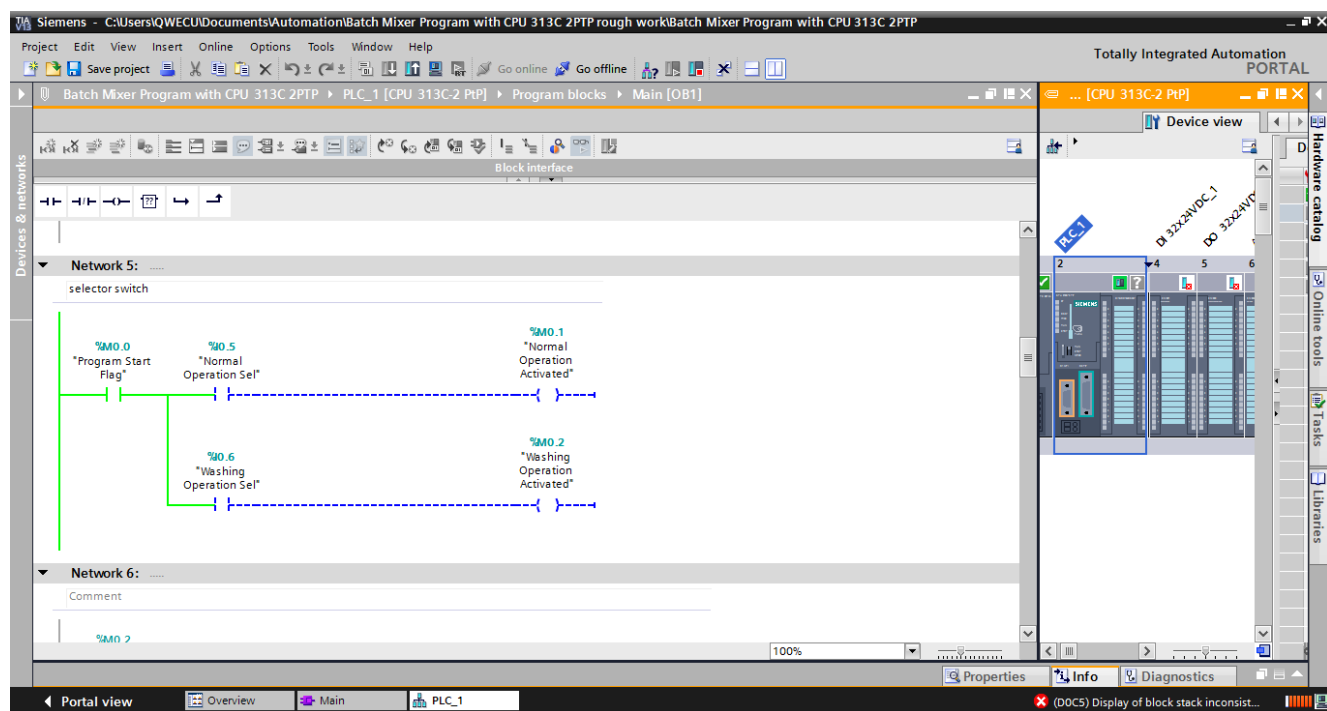


Figure A3. Optional Selection of Operation.

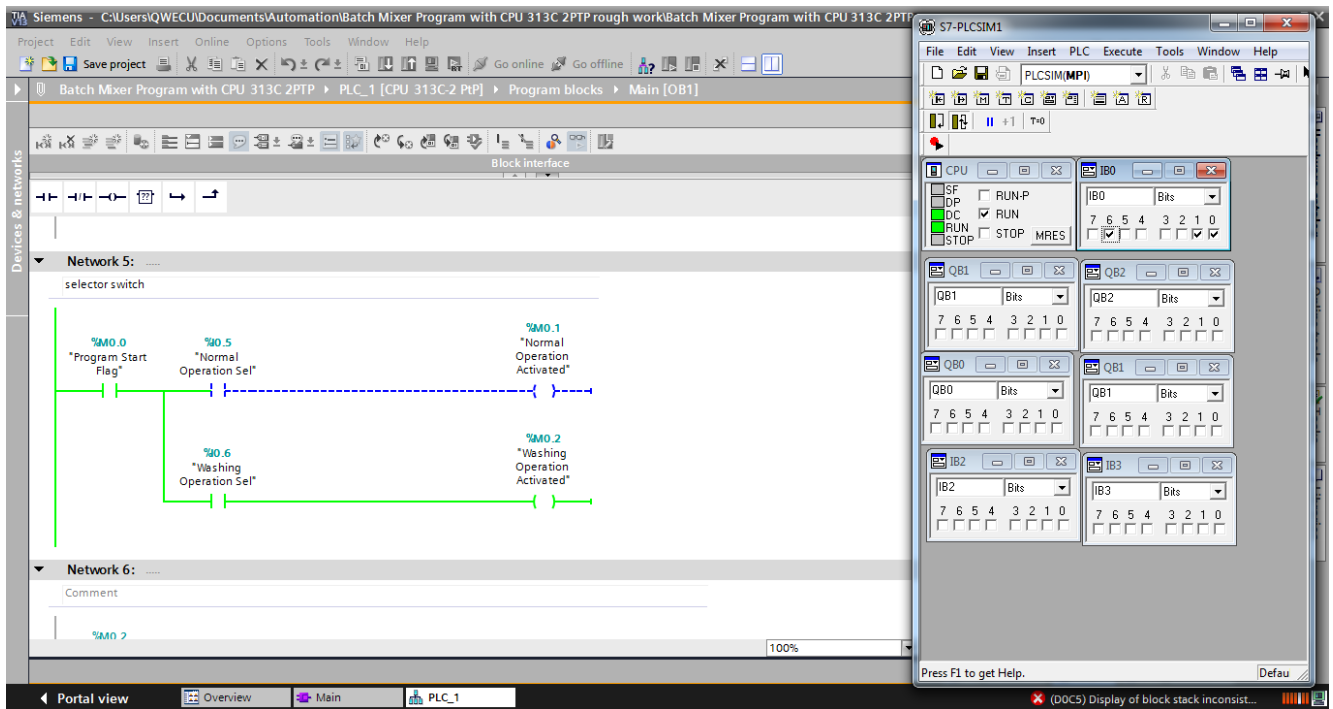


Figure A4. Washing Operation Selected.

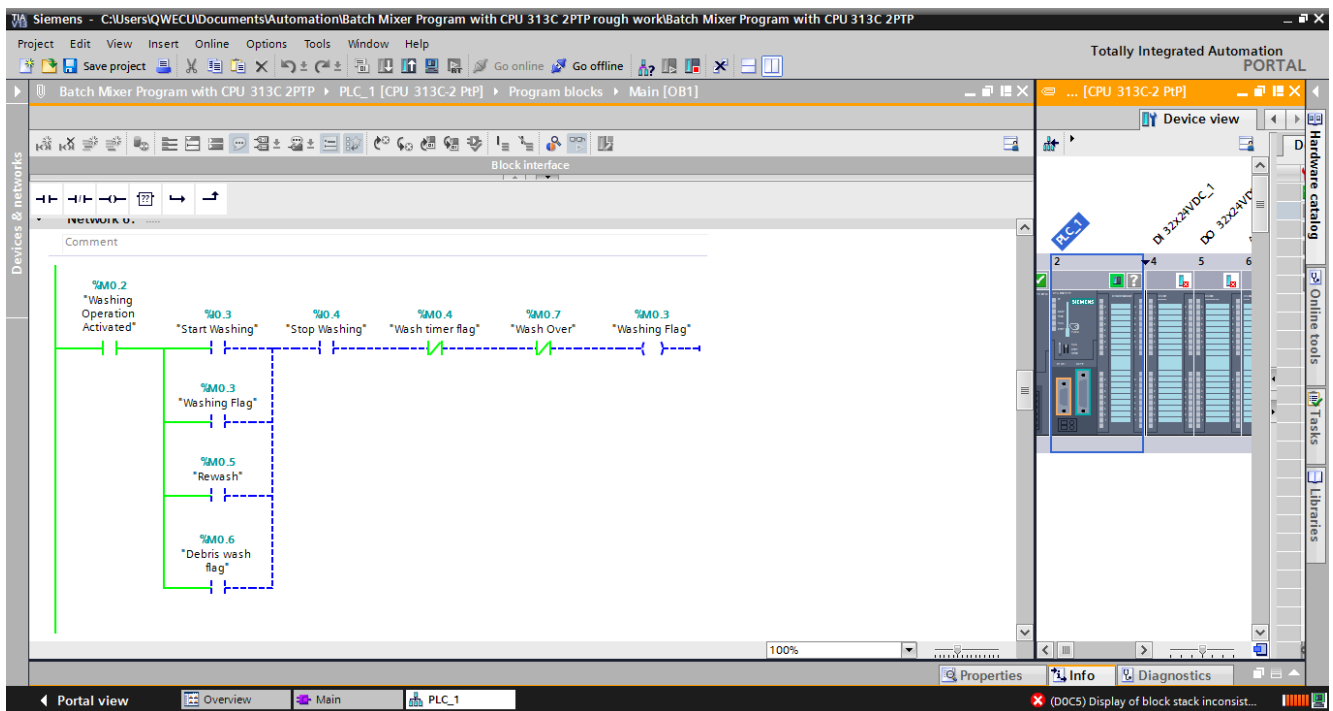


Figure A5. Activation of Push Button for Washing.

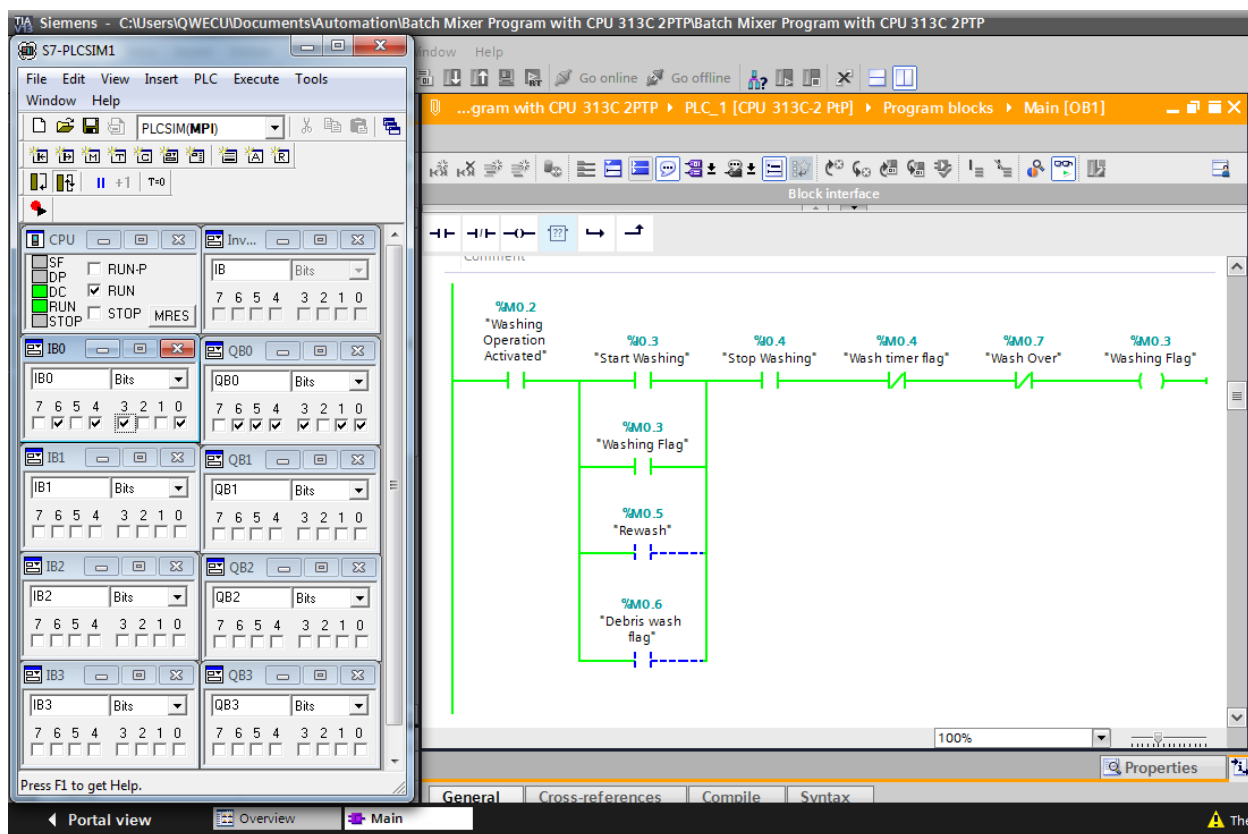


Figure A6. Washing Sequence Activated.

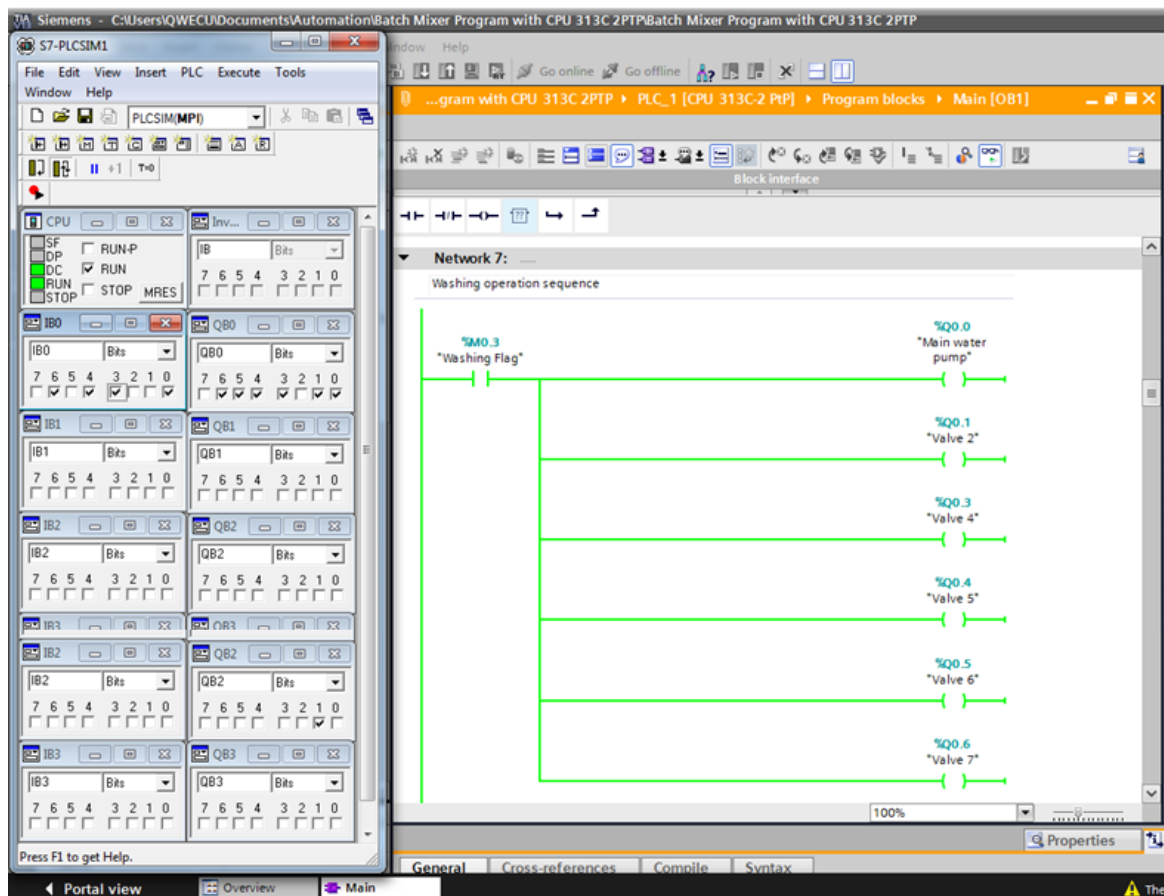


Figure A7. Inlet Valve Open for Washing.

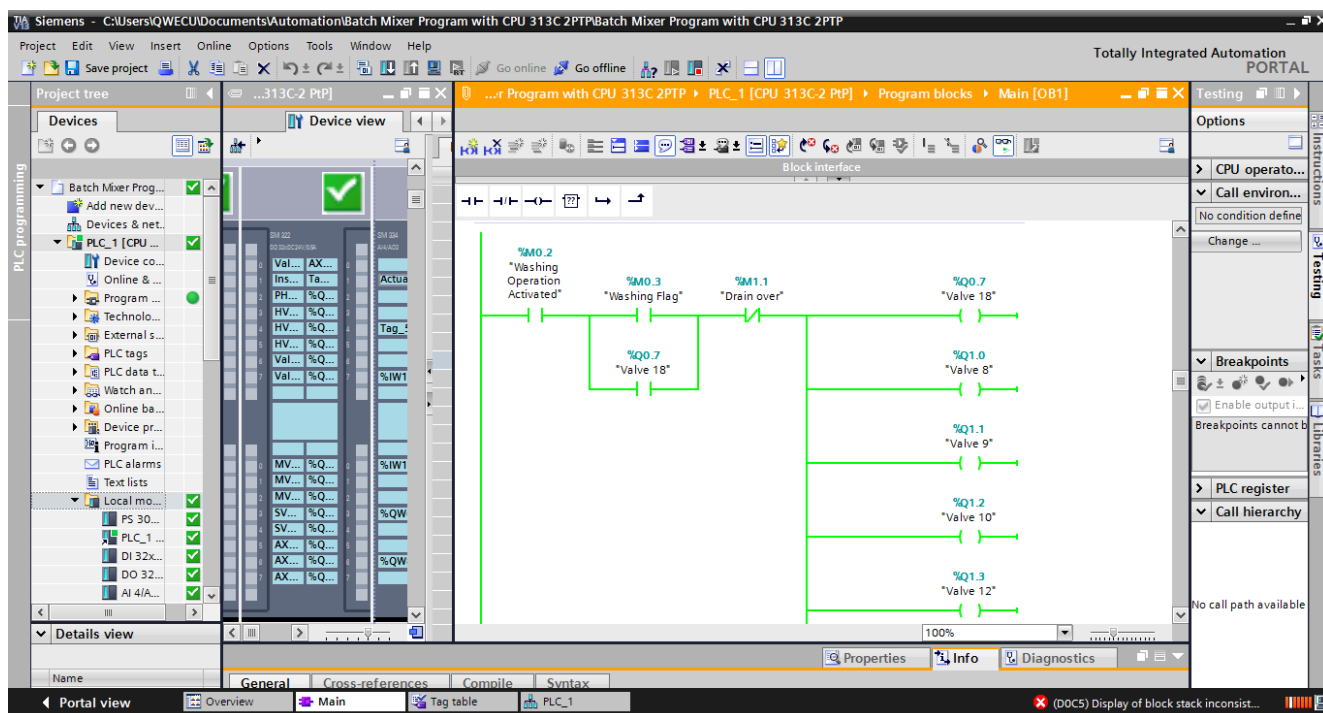


Figure A8. Outlet Valves 8-12 Open for Washing.

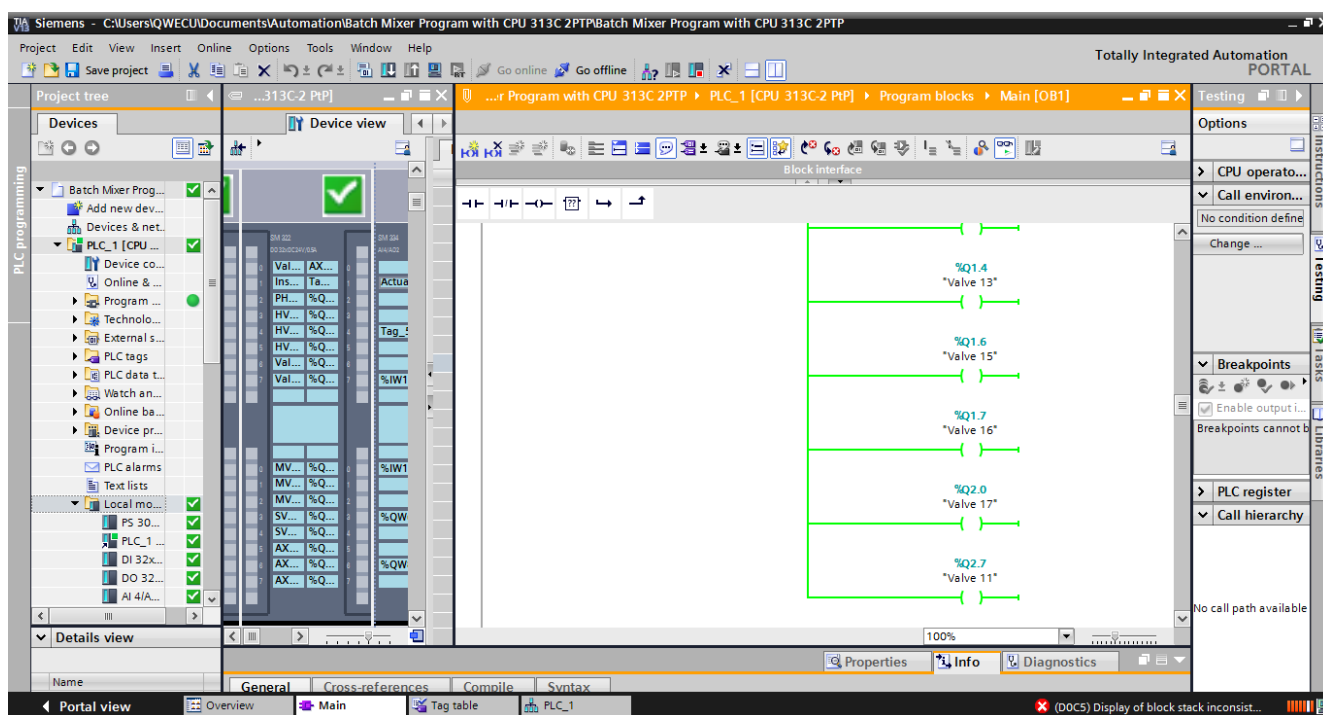


Figure A9. Outlet Valves 11, 13-17 Open for Washing.

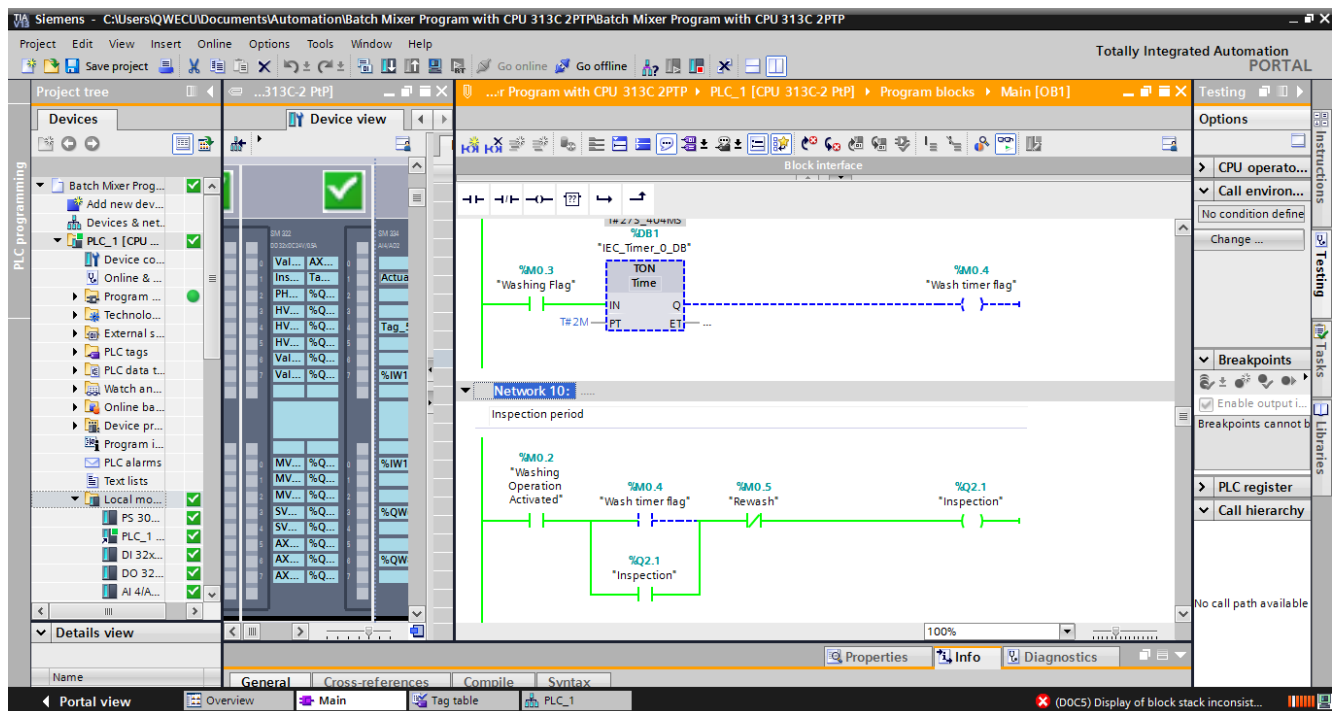


Figure A10. Washing Timer Elapses and Inspection of Vessel Starts.

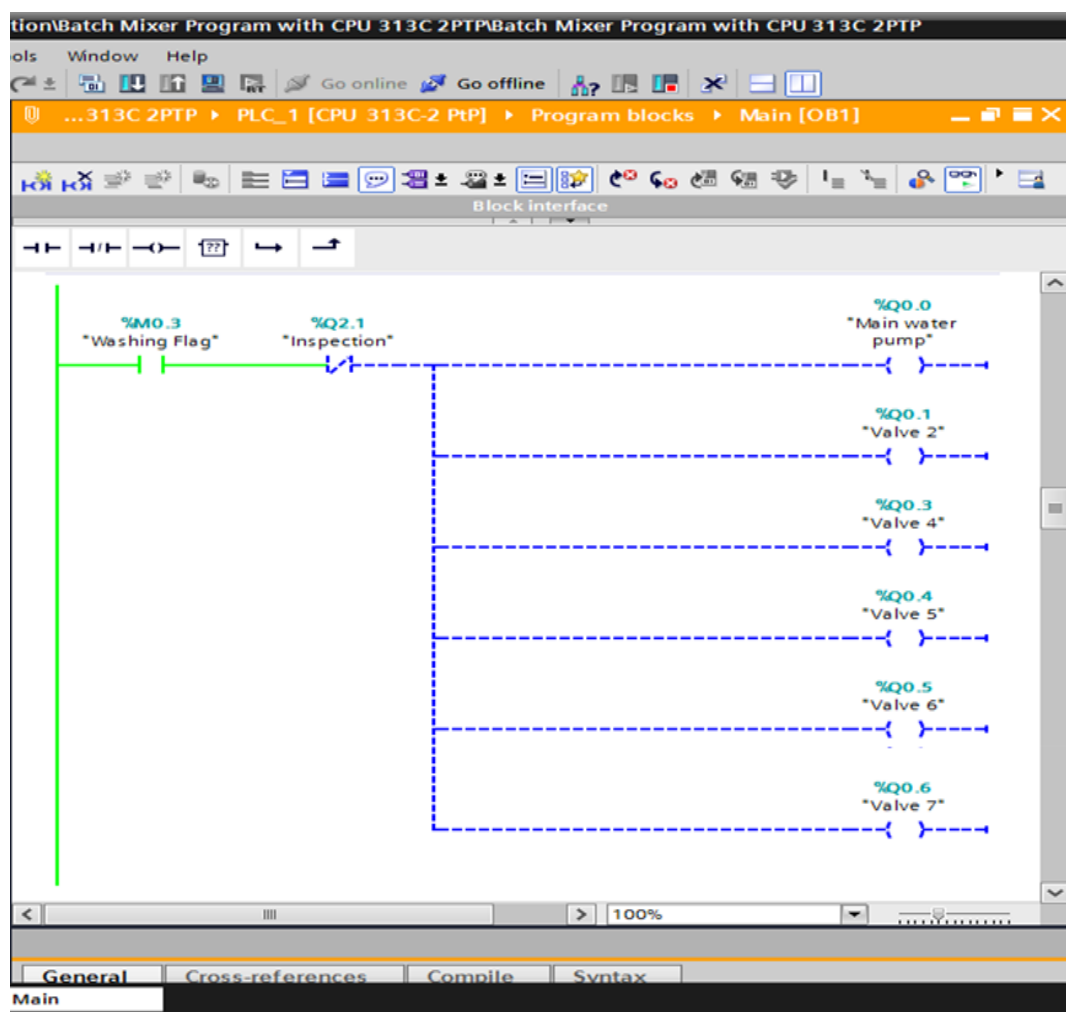


Figure A11. Inlet Valve Close for Inspection.

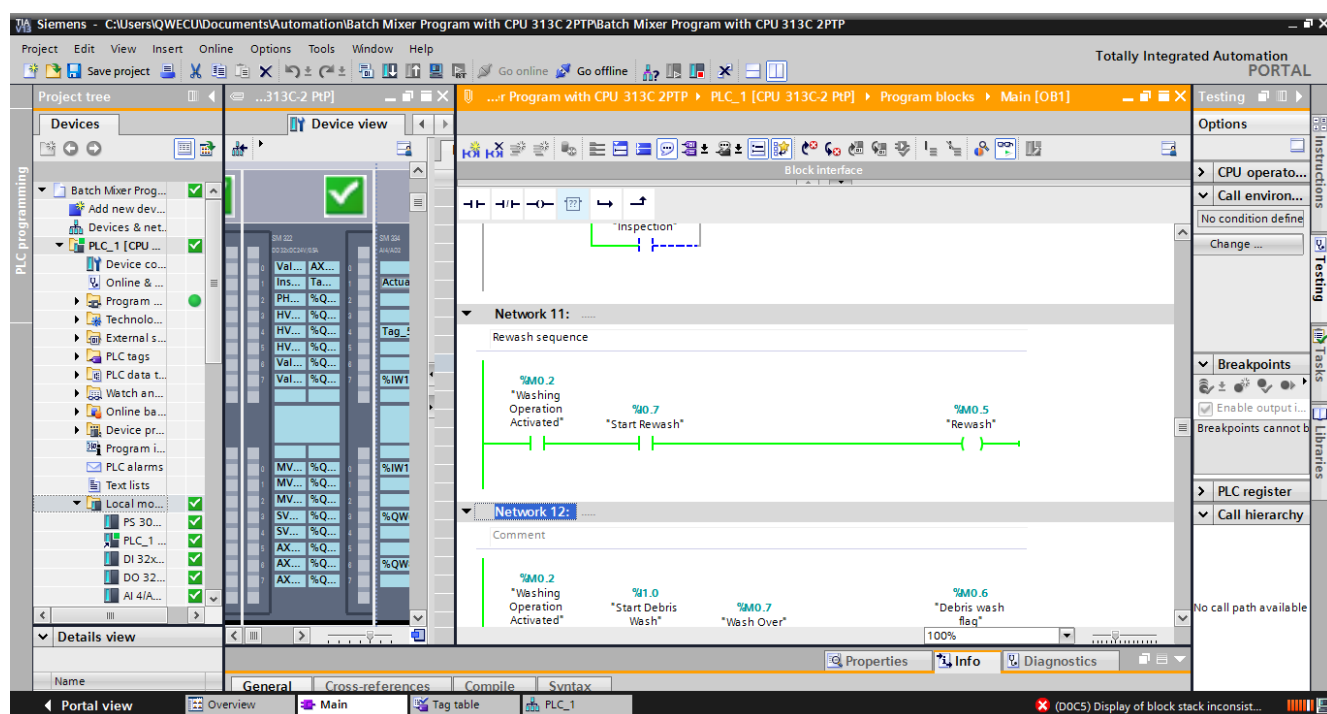


Figure A12. Rewash Activated.

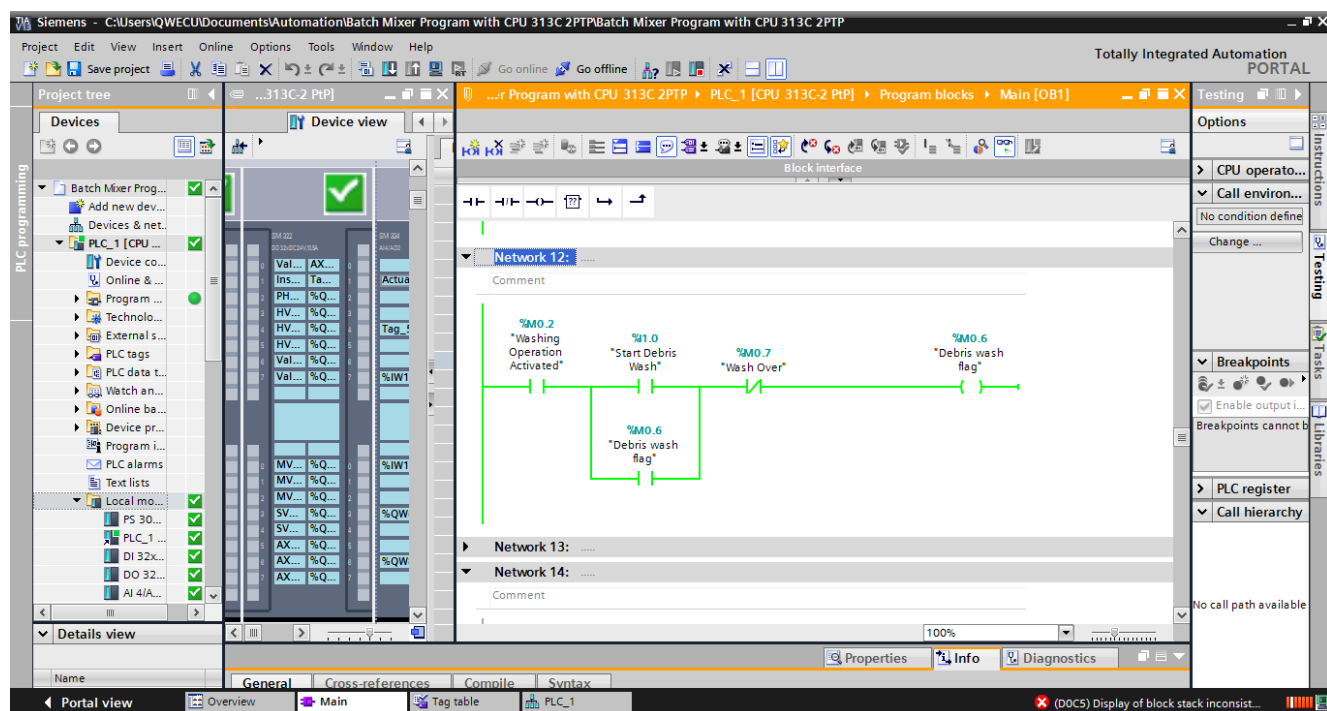


Figure A13. Debris Wash Activated.

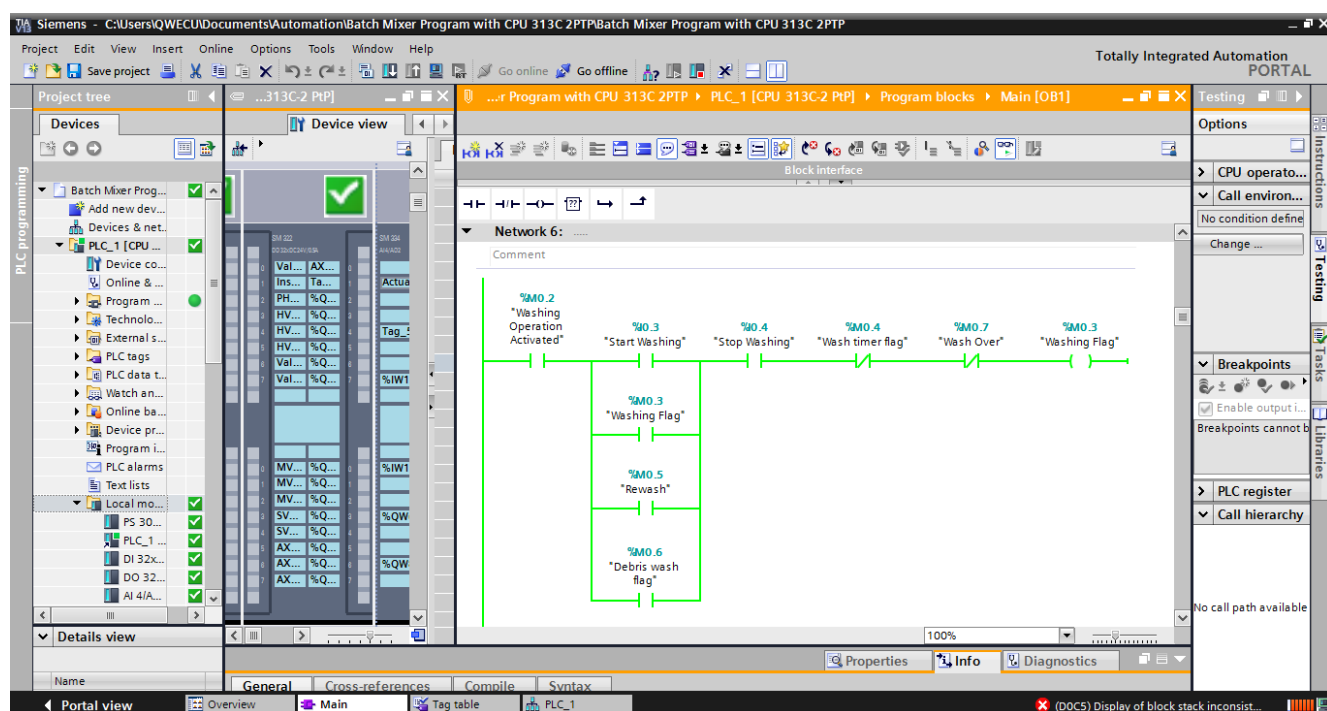


Figure A14. Final Wash After Inspection of Debris.

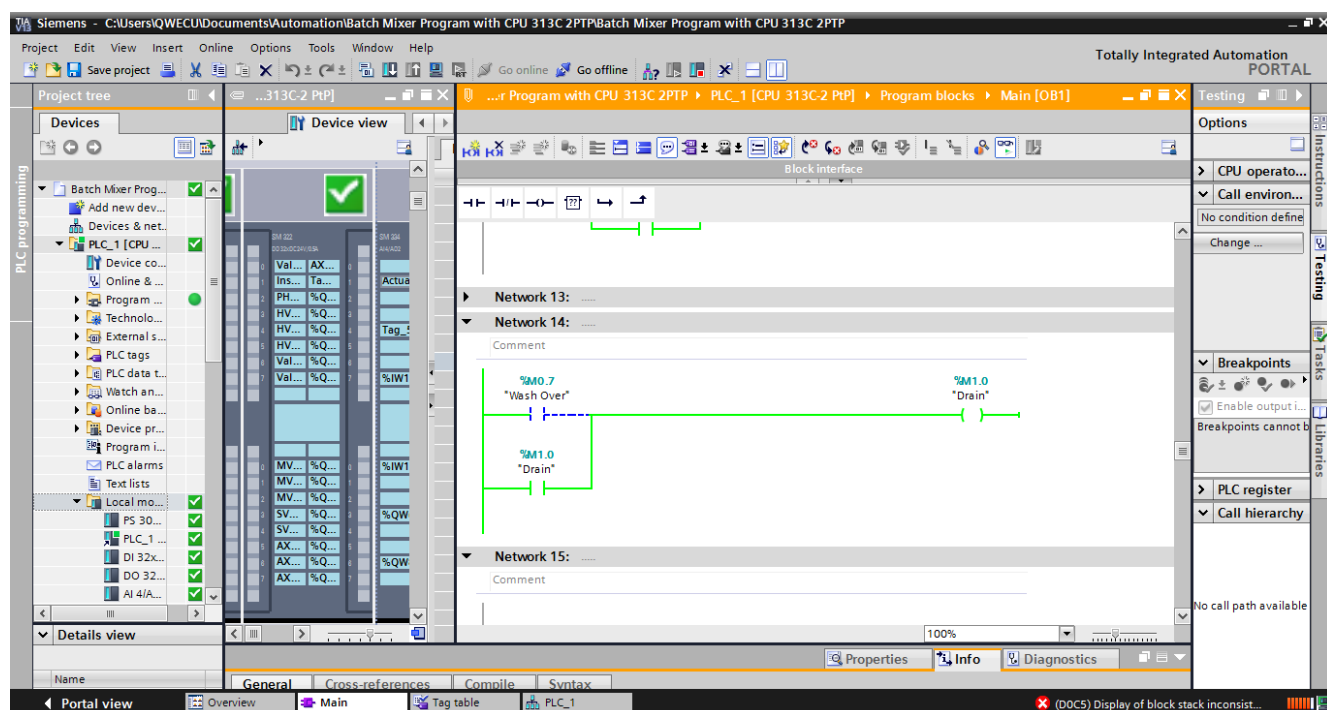


Figure A15. Washing Over and Time Due for Drain.

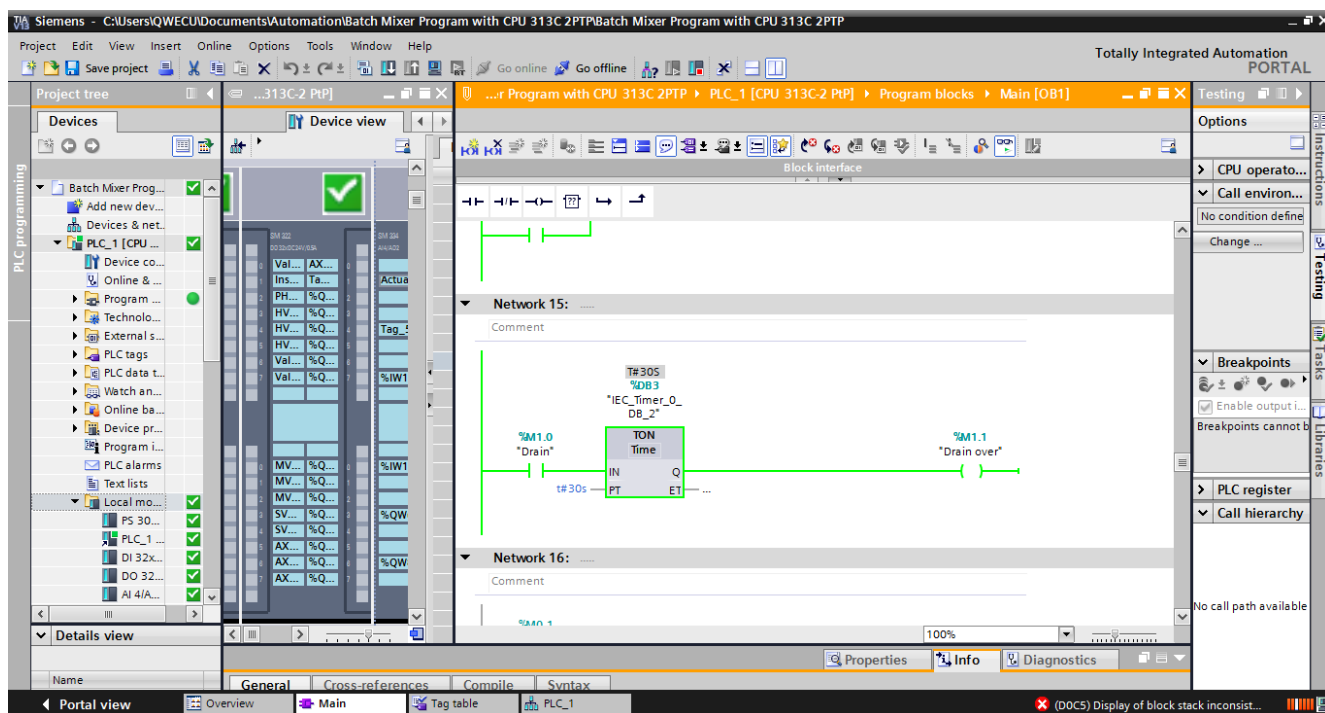


Figure A16. Drain Time Elapses for Draining.

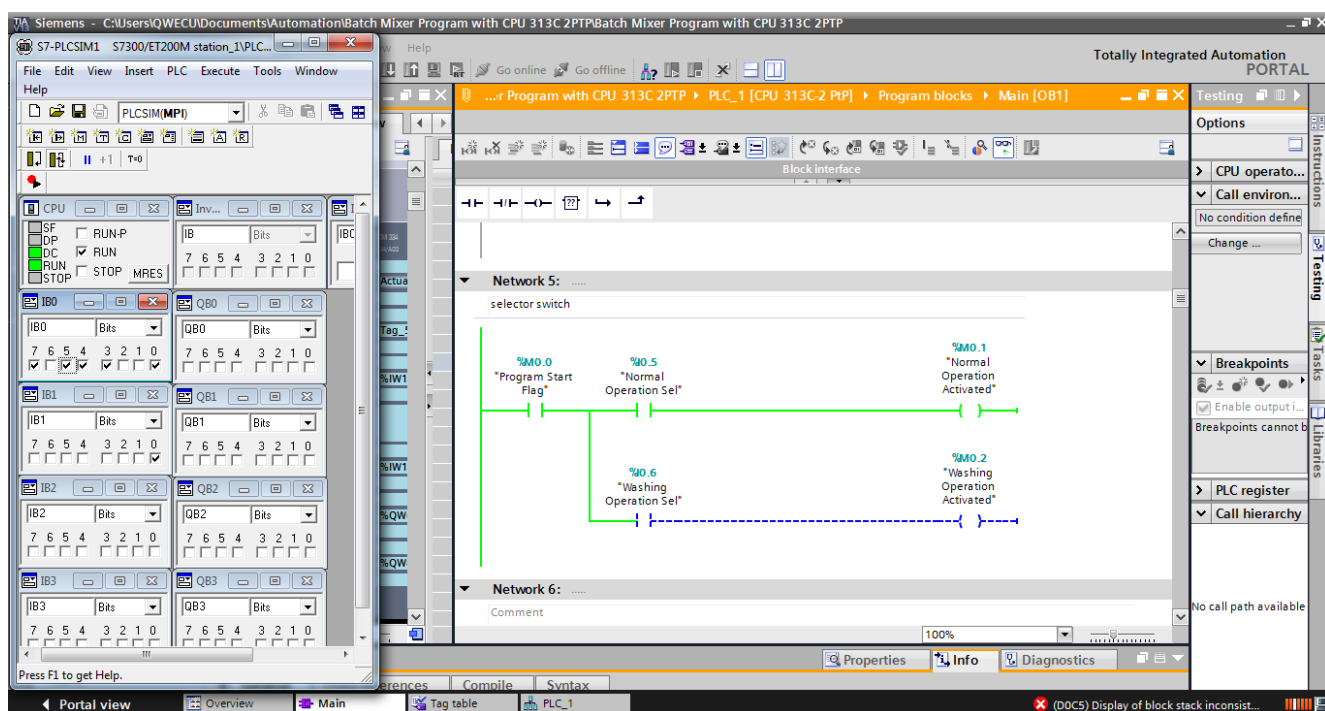


Figure A17. Batch Operation Selected.

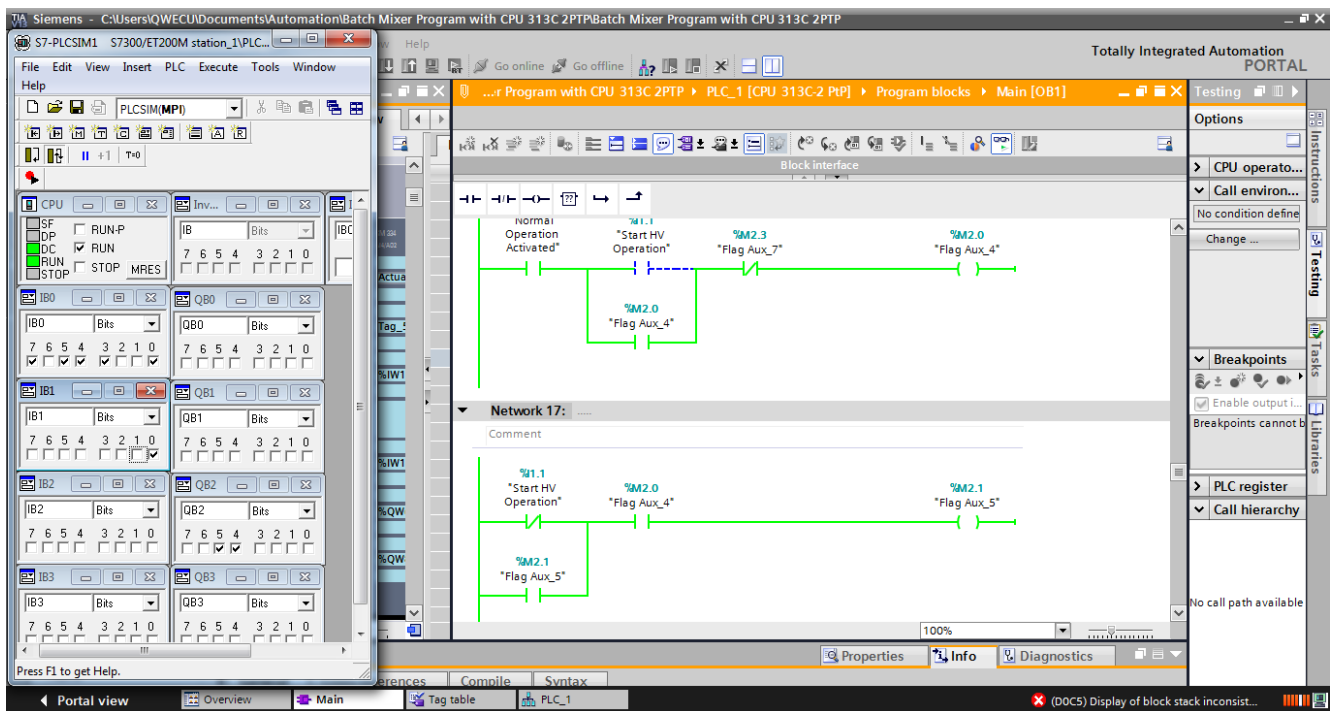


Figure A18. Start-Stop Activation of Batch Operation.

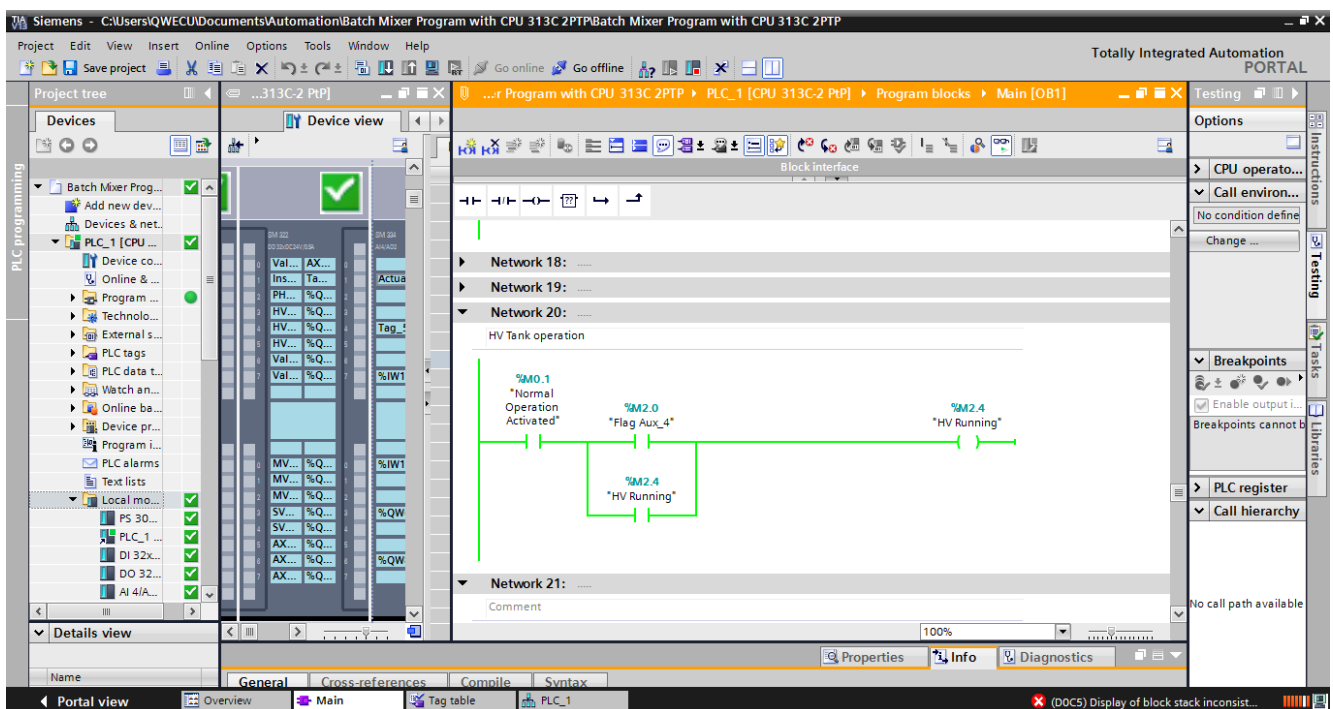


Figure A19. Heating Tank Operation.

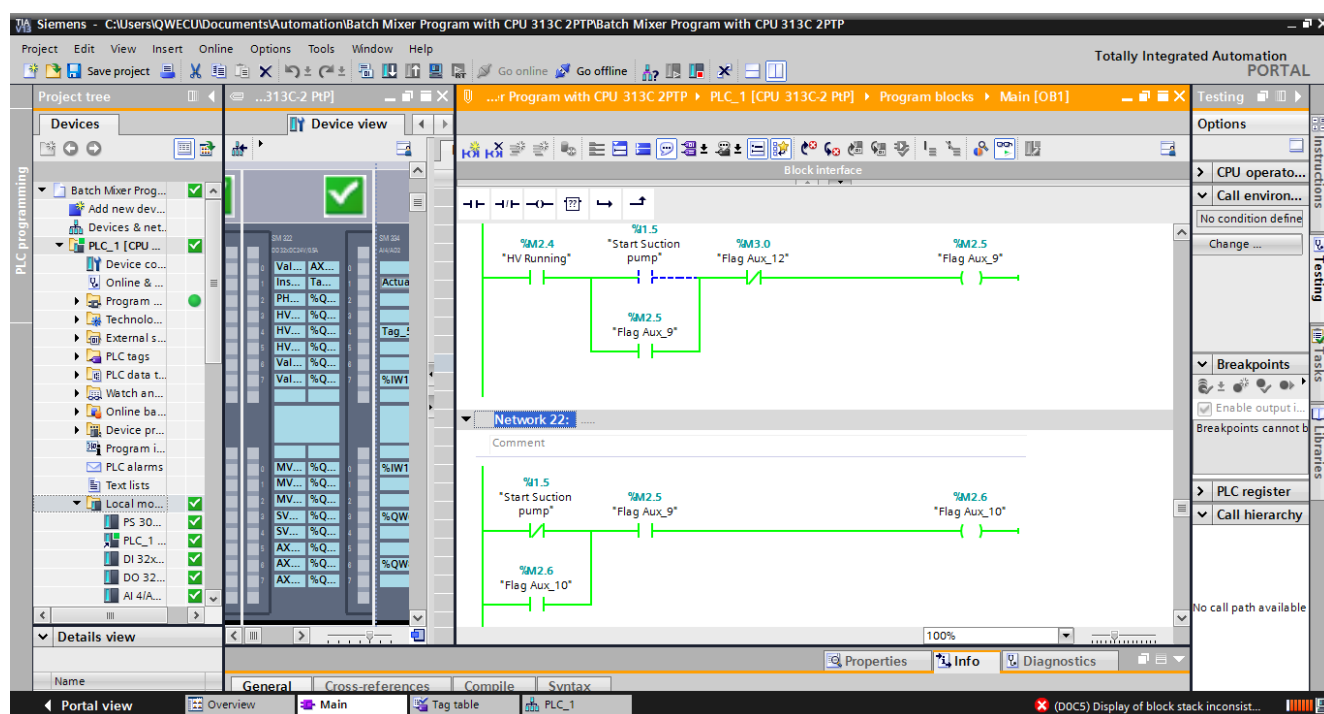


Figure A20. Suction Pump is Started.

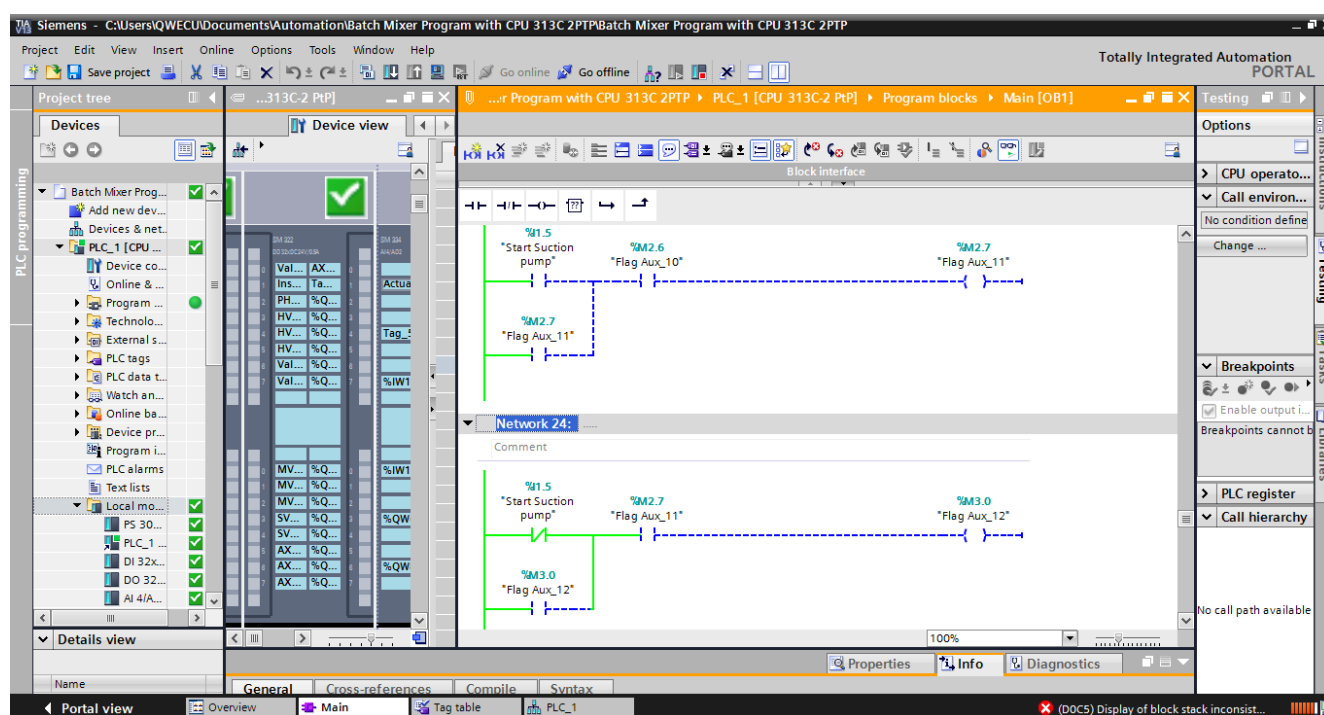


Figure A21. Suction Pump Stopped.

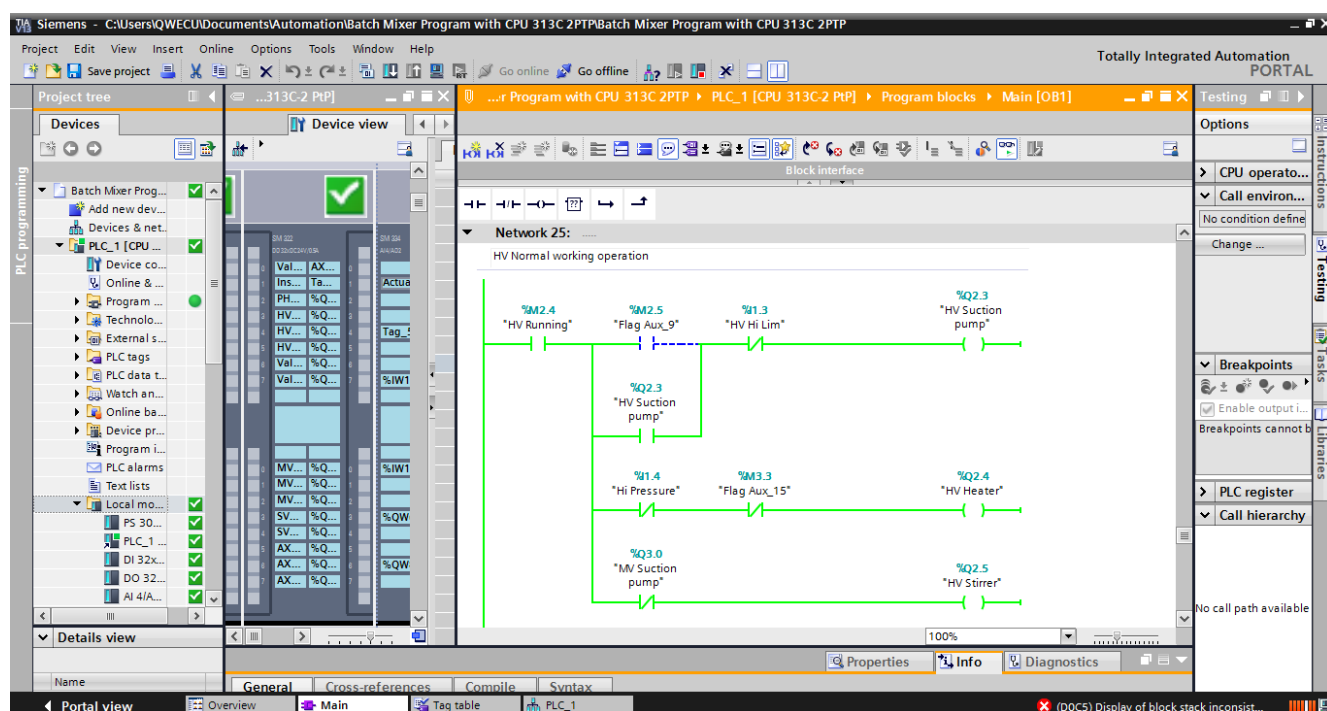


Figure A22. Heating Tank Agitator and Heaters Starts.

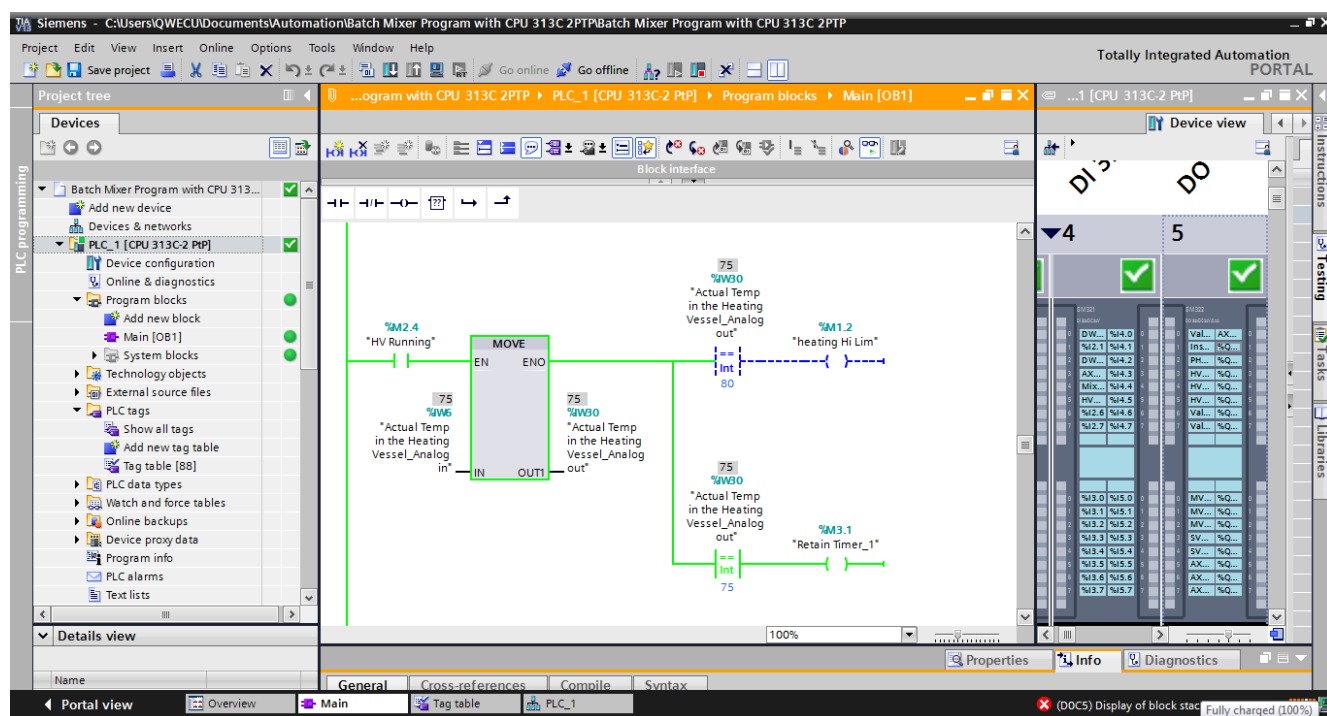


Figure A23. Content Heated at a Temperature of 75 °C.

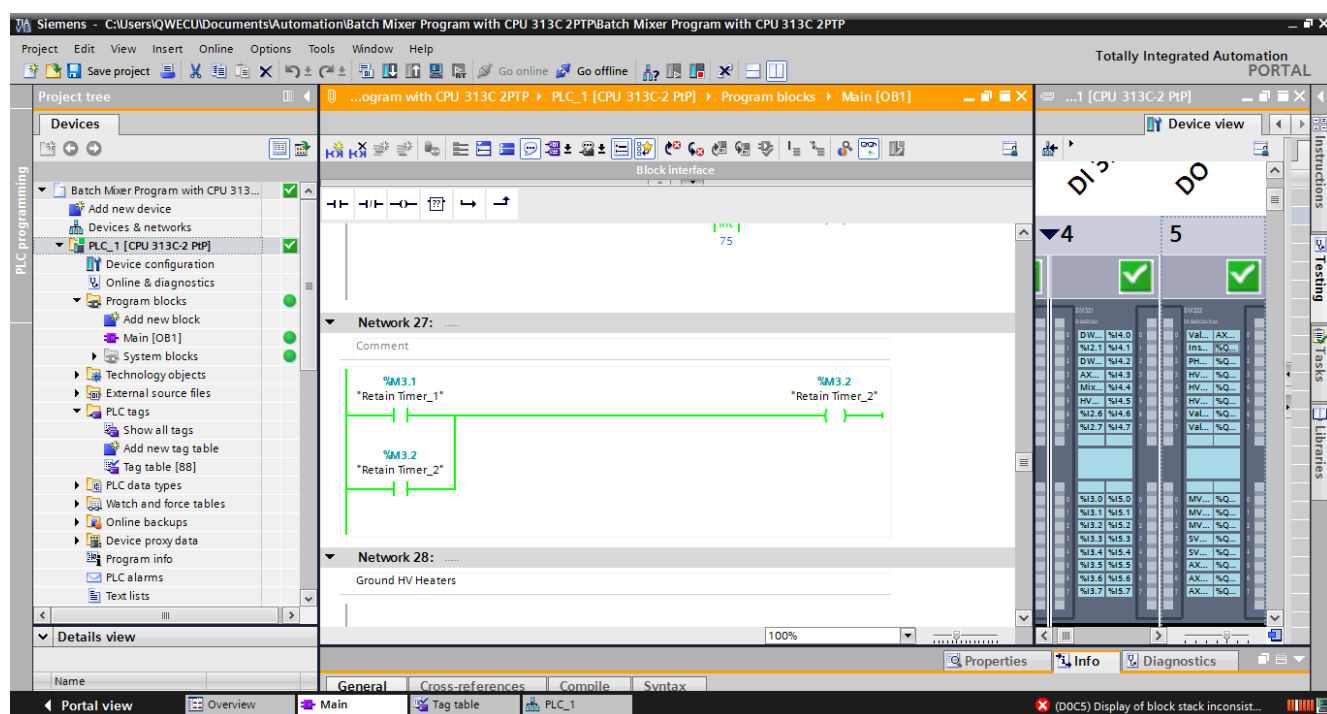


Figure A24. Heating Vessel Timer is Retained ON.

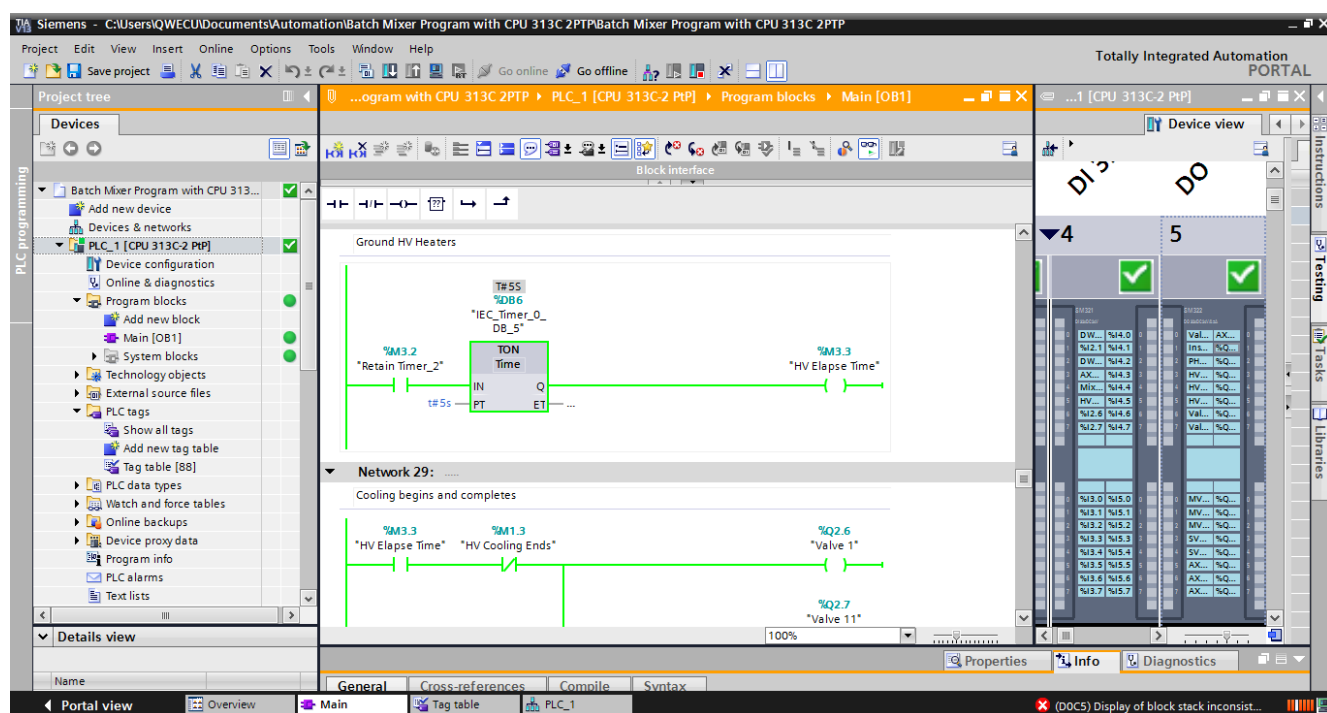


Figure A25. Heating Time Elapsed.

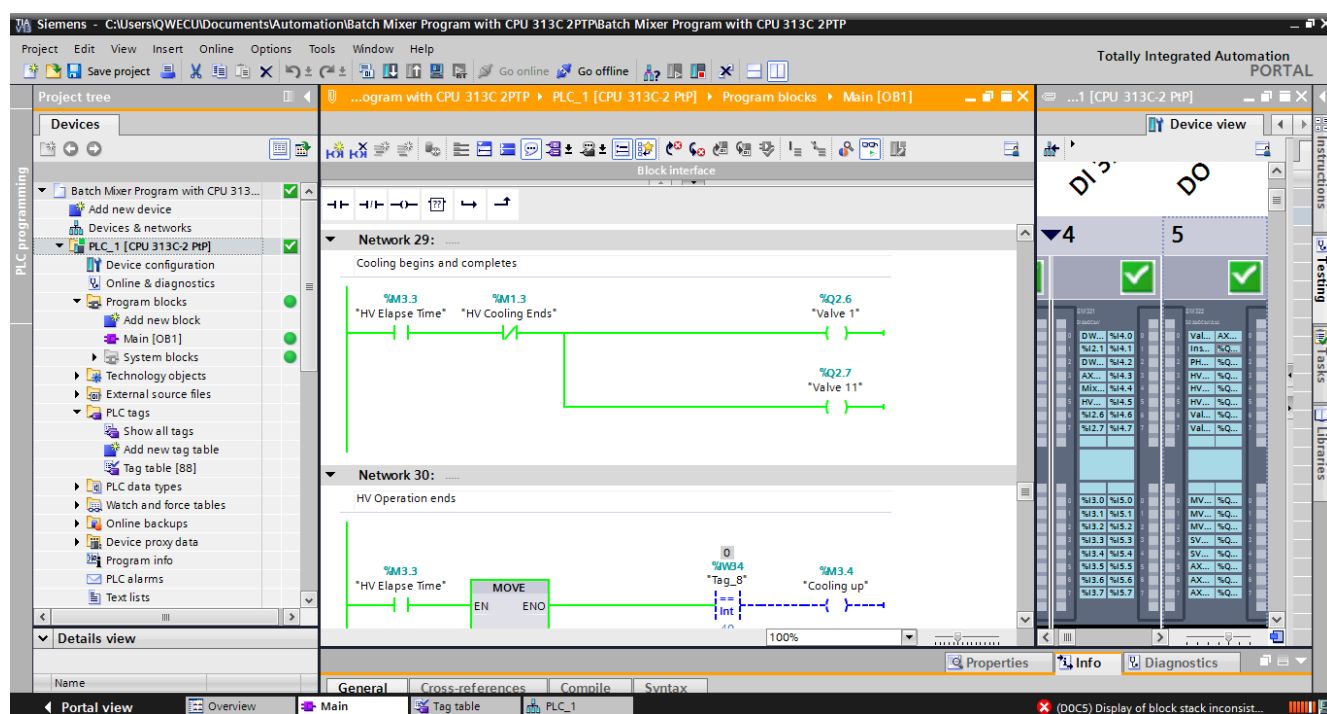


Figure A26. Activation of Cooling Valves When Heating Time Elapses.

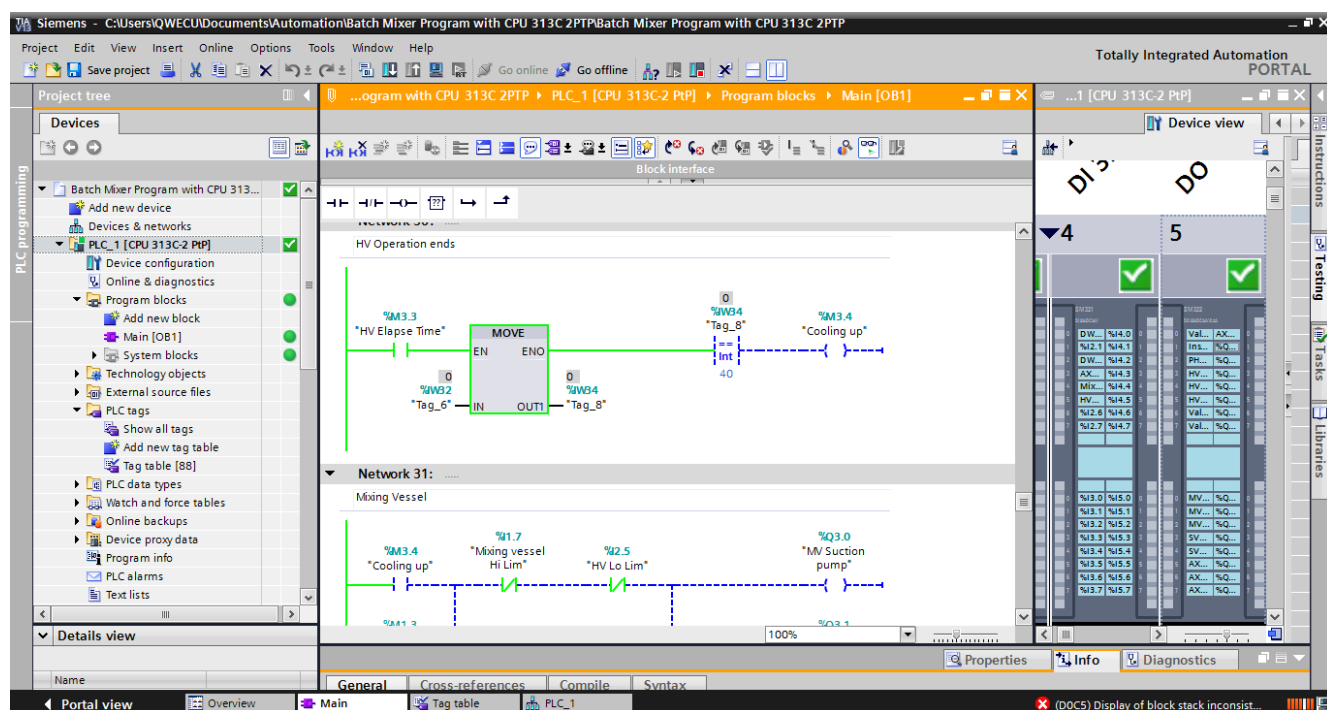


Figure A27. Activation of Cooling Timer.

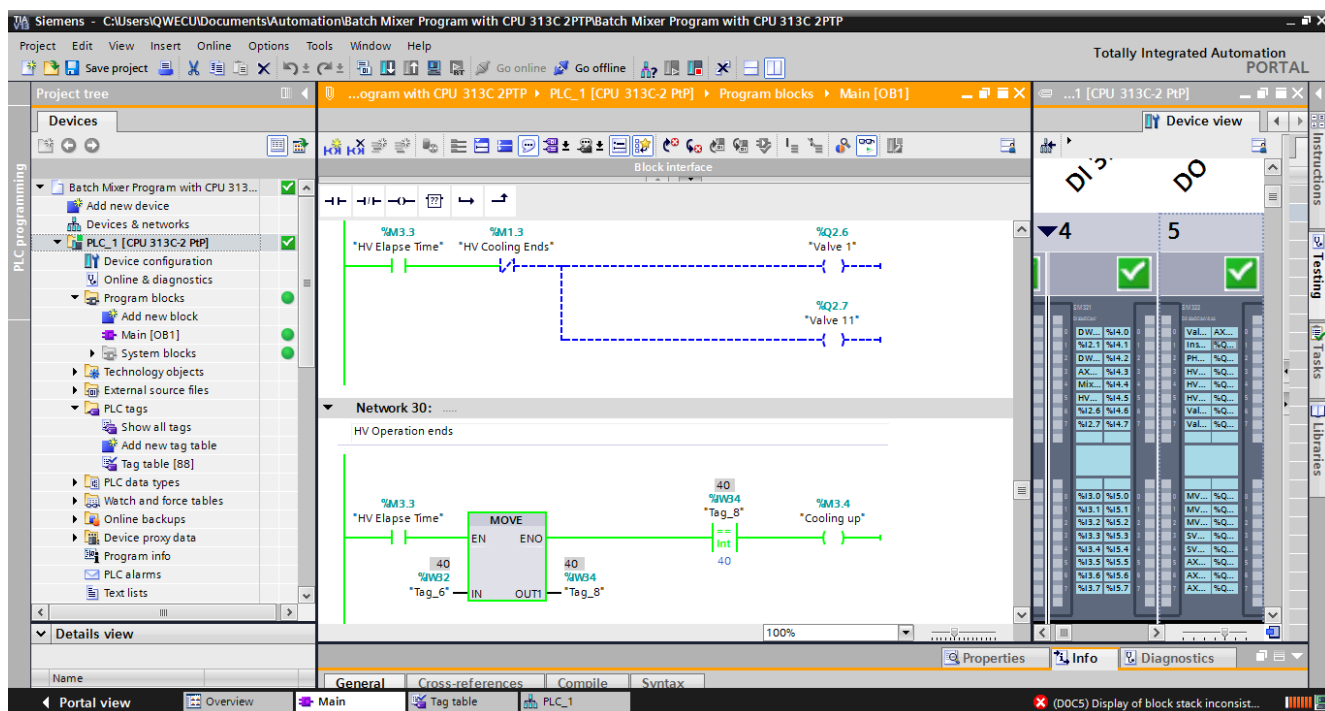


Figure A28. Deactivation of Cooling Valves at Elapsed Time.

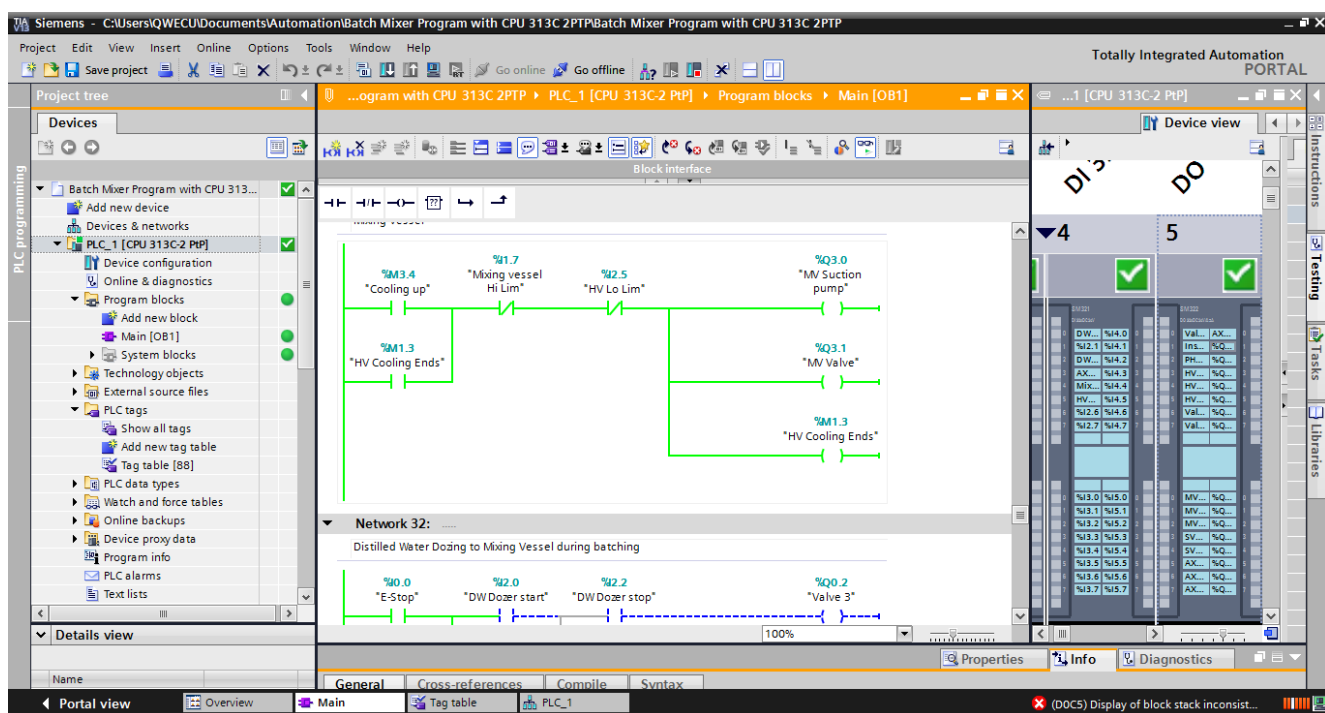


Figure A29. Mixing Tank, Suction Pump P1 Starts at the End of Cooling.

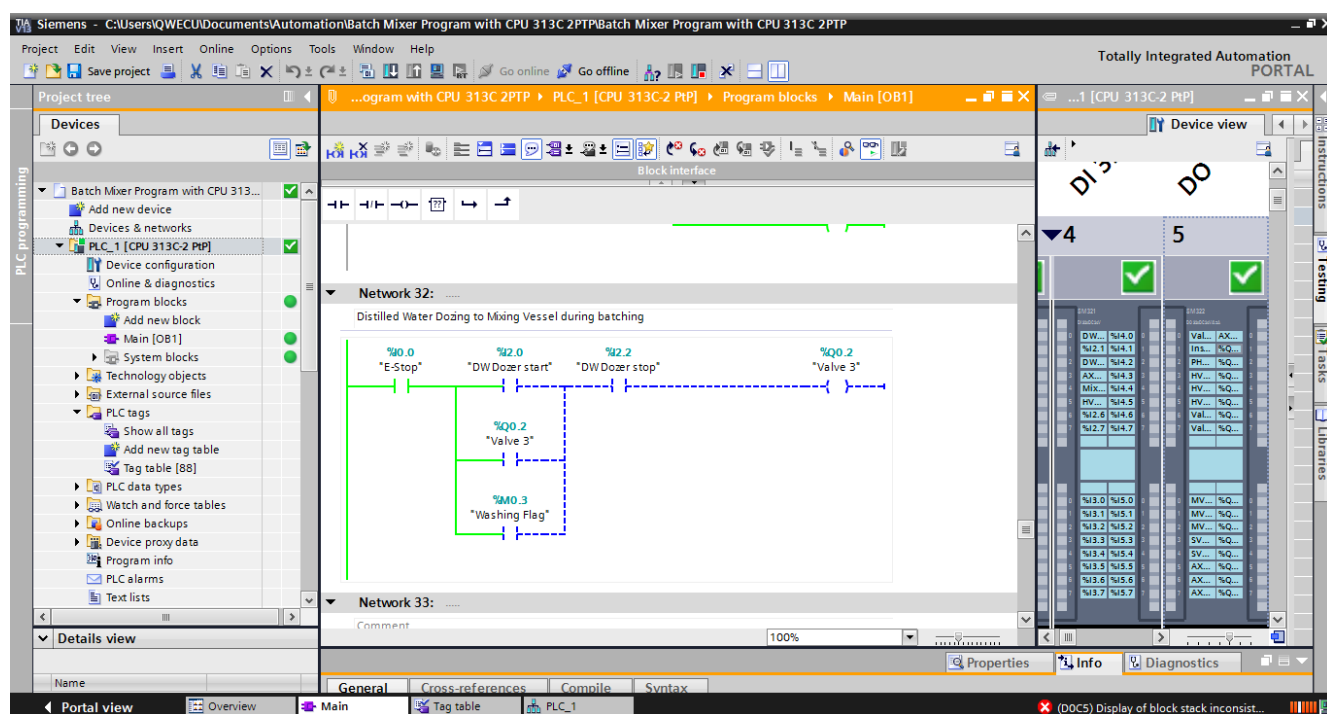


Figure A30. Activation of Distilled Water for Mixing the Batch.

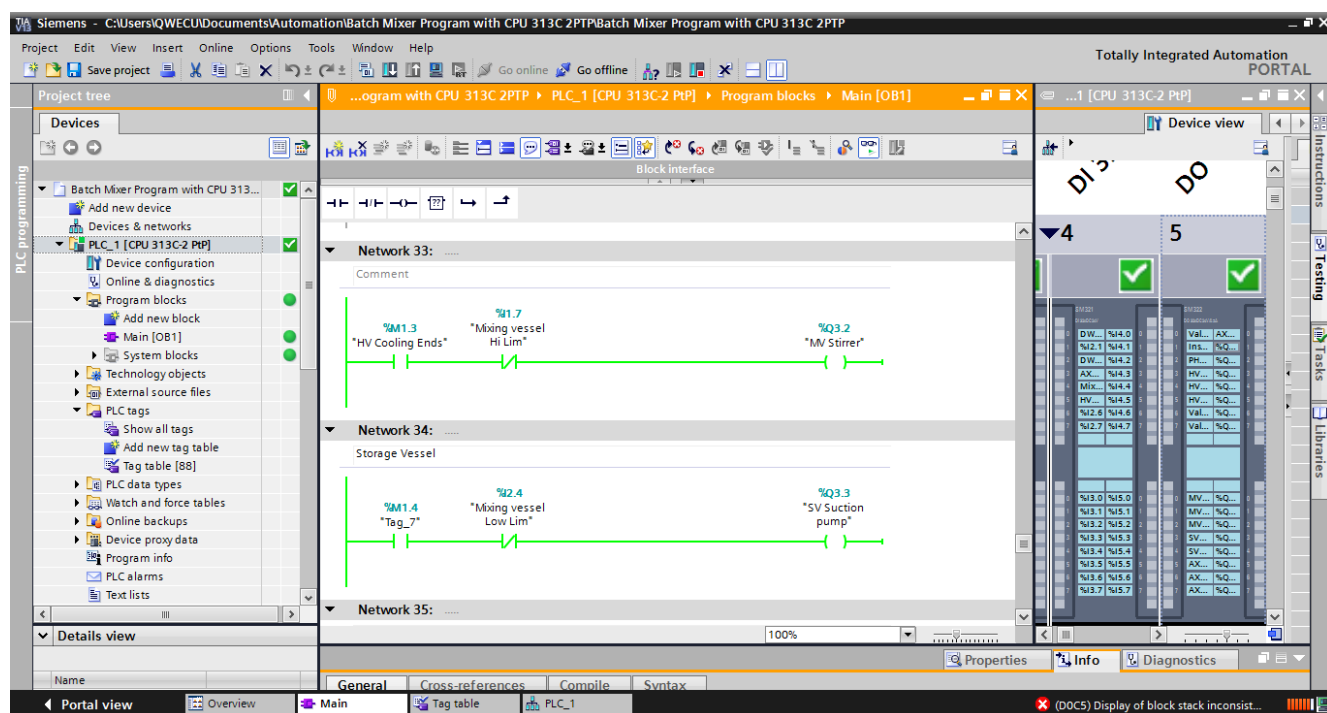


Figure A31. Mixing Tank Agitator Starts.

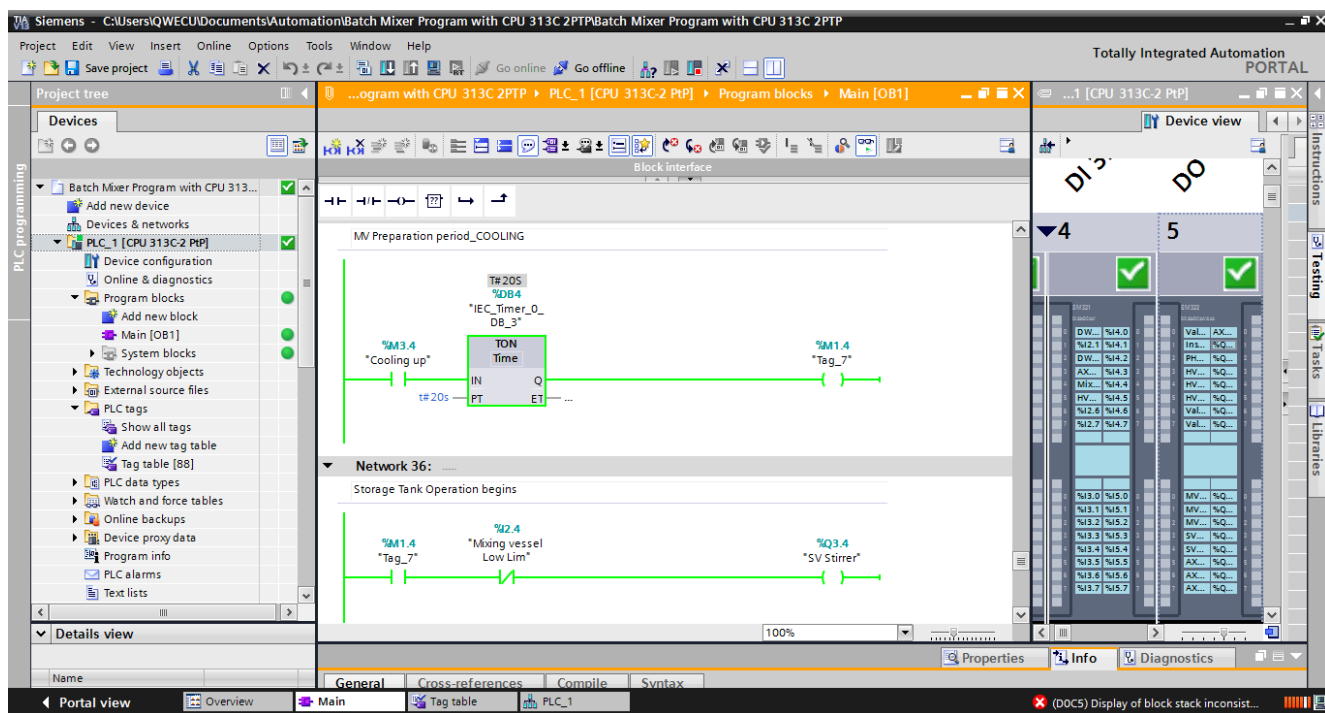


Figure A32. Storage Tank Agitator Start at Low Limit Mixing Tank.

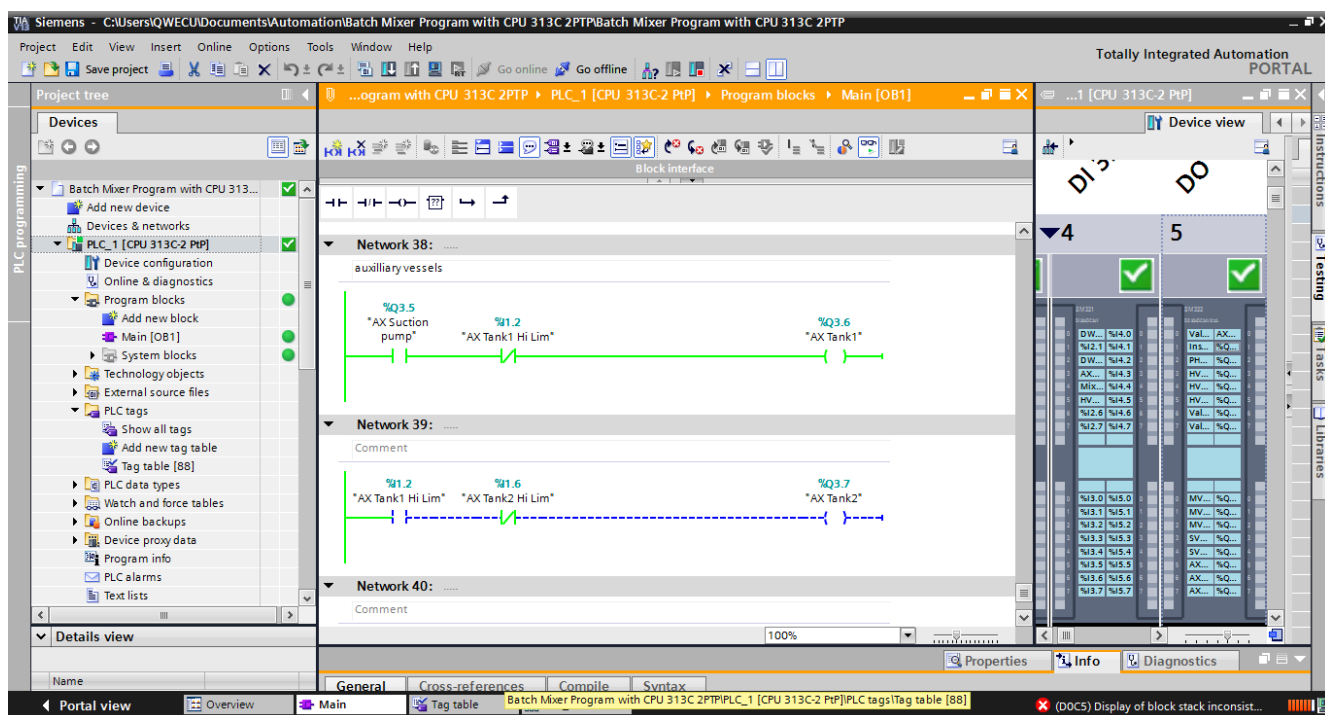


Figure A33. Filling Auxiliary Tank 1.

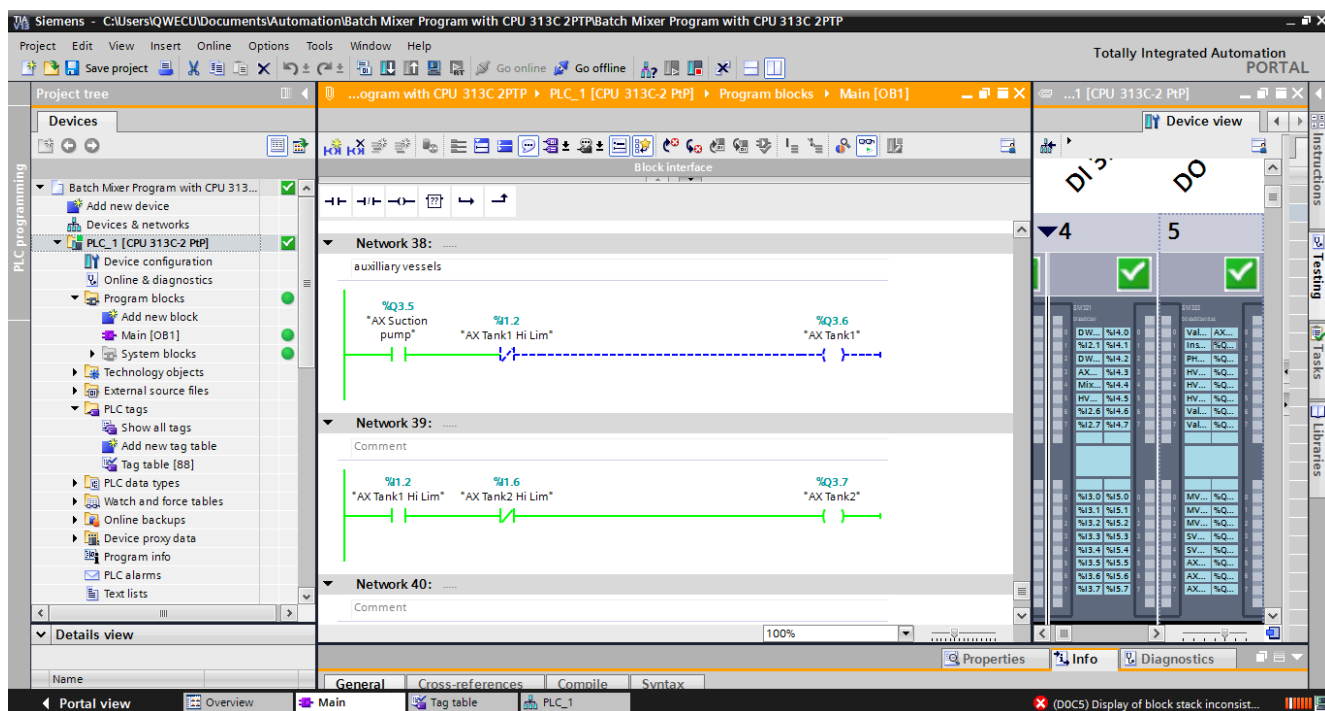


Figure A34. Filling Auxiliary Tank 2.

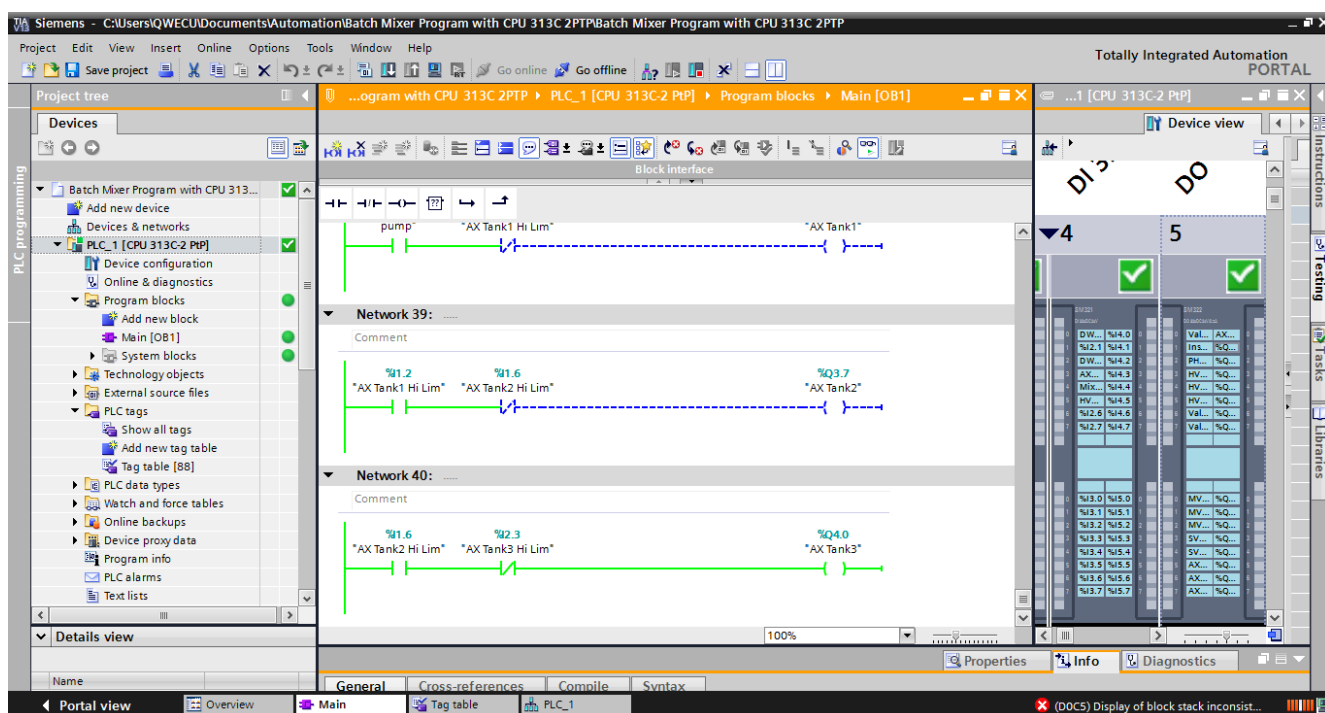


Figure A35. Filling Auxiliary Tank 3.

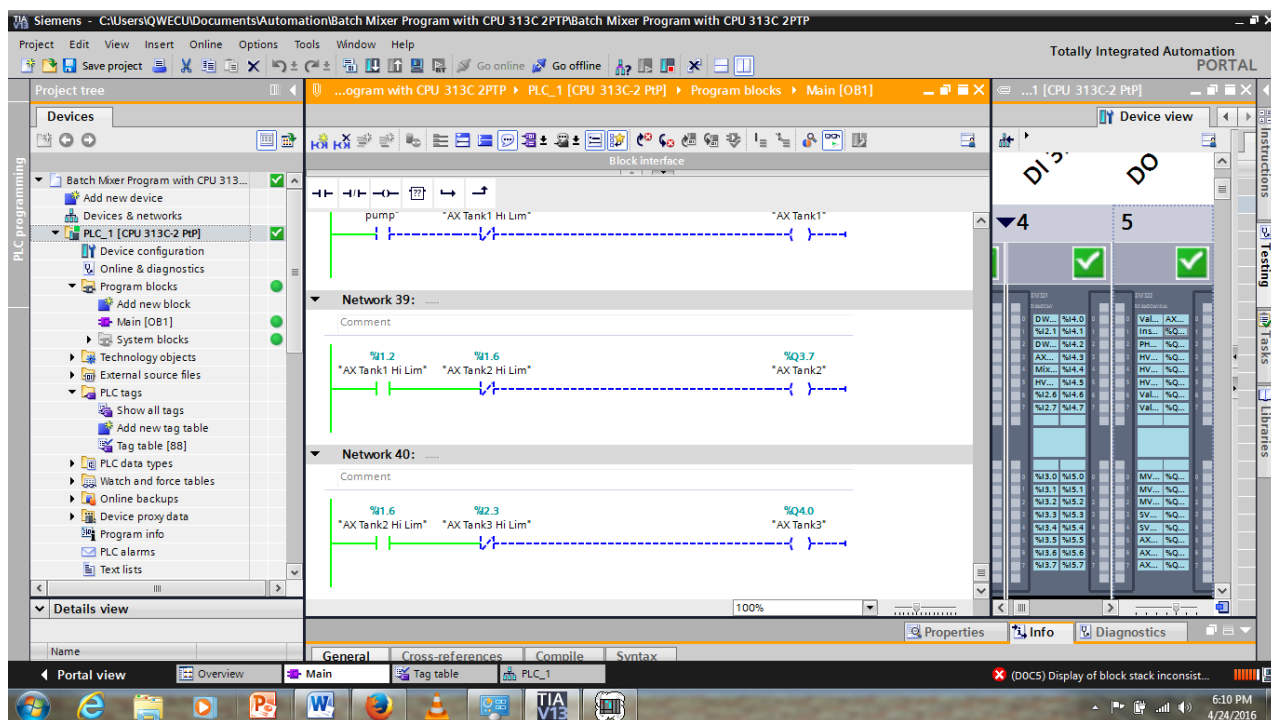


Figure A36. Auxiliary Tank 3 High.

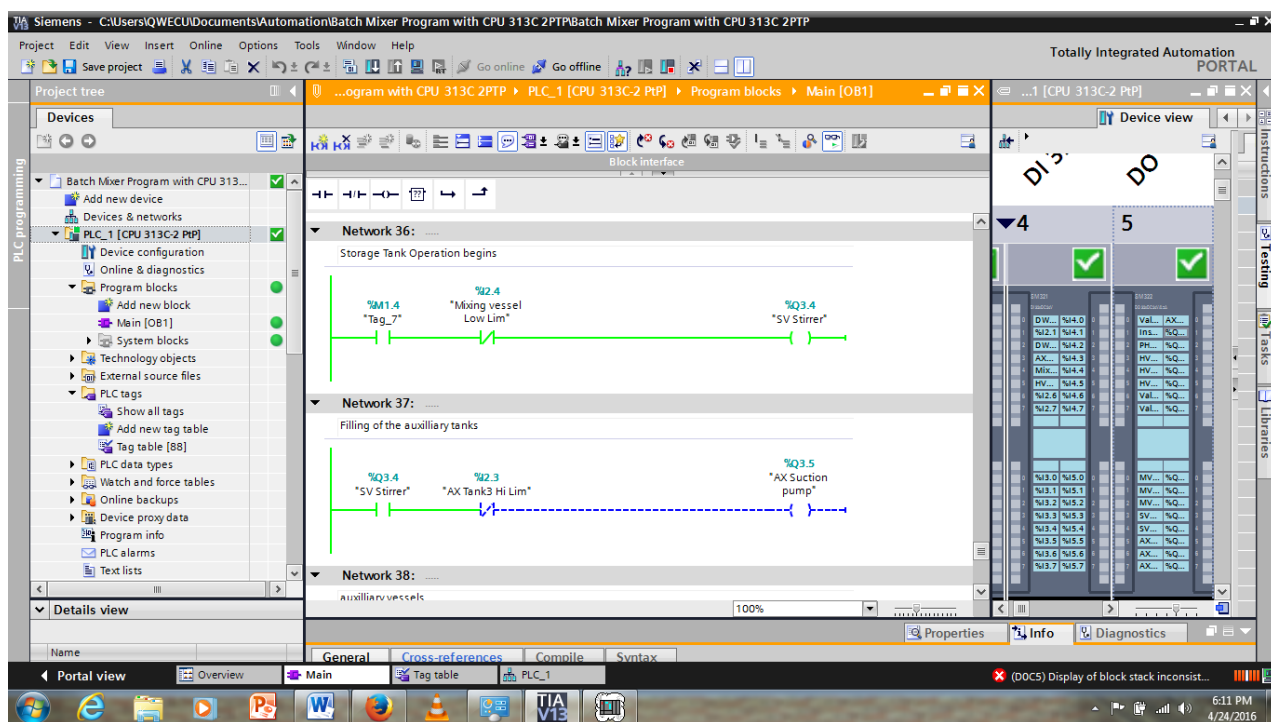


Figure A37. Auxiliary Suction Pump Stops.

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